A signal-to-drift ratio for quantifying winding-fault detectability and selecting condition monitoring across three wind-turbine generator types

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Abstract

Detection methods for incipient stator-winding faults are developed one machine at a time, so how the generator type changes the problem is not quantified, and the detectability measures of fault-detection theory presuppose a model or a trained detector. A statistic that requires neither is proposed, the signal-to-drift ratio (SDR): the relative change of a feature between a healthy reference interval and a faulted interval, divided by a quantile of its changes between adjacent healthy intervals. The SDR is the known-onset mean-shift test as a standardised effect size, calibrated on each machine's own healthy non-stationarity and comparable across features, operating points and machines. From it follow a minimum detectable extent, a detectable share and a signature location per machine type, sensor suite and fault class; a decision table selects the cheapest suite that meets a requirement with a stated power and bootstrap confidence; and a transfer test fixes the feature-referencing rule under which a detector serves several machine types. It is demonstrated on open laboratory-bench data from permanent-magnet, squirrel-cage induction and wound-field synchronous generators with the same tapped inter-turn and inter-winding short-circuits. The same design-level fault is three different physical events; phasor residuals remove a faulted-phase dependence of the magnitude ratios; the signature relocates between terminal voltage, controller states and field current with the machine–drive combination; the cheapest adequate suite depends on machine type and requirement; and a

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learned detector transfers once its features are referenced to the target machine's healthy behaviour. Results carry bootstrap intervals; code and data are open.

Keywords: condition monitoring, fault detectability, stator winding fault, wind-turbine generators, sensor suite selection, transfer across machines

1. Introduction

Stator-winding insulation faults are the archetypal incipient failure of an electrical machine: a short-circuit across a few turns with subtle signatures that evolves into a machine-destroying fault if it is not detected [1, 2]. In wind turbines, electrical-subsystem failures contribute substantially to downtime [3, 4], and a rich literature detects winding faults in a given machine: motor-current signature analysis [5], negative-sequence current and impedance methods [6, 7], the review literature [8, 9] and, for converter-fed machines, the analysis of how closed-loop current control moves the asymmetry from the currents into the voltage commands [10, 11]. Almost all of it is developed and validated on one machine. Yet the generator type of a large turbine—permanent-magnet synchronous (PMSG), squirrel-cage induction (SCIG), wound-field synchronous (WFSG) or doubly-fed induction—is itself chosen on cost, mass, efficiency and grid-compliance grounds [12, 13], and its condition monitoring is added afterwards from a catalogue of commercial systems [14]. Whether the same fault is equally visible in different generator types, in which measured quantity it appears, and which sensors are therefore worth specifying for each has not been quantified.

Answering such questions requires a measure of detectability with three properties that the existing measures do not combine. It must be computable without a model of the machine and without a trained detector, because the quantity of interest is what the measurements contain, not what one algorithm extracts from them; the detectability and isolability notions of fault detection and isolation (FDI) theory [15–17] and the sensor-placement methods built on them [18] presuppose a model, and a learned detector [19] presupposes data from the very machine, with transfer across machines an open problem [20]. It must be comparable across features in different physical units, across operating points and across machines whose healthy behaviour differs, which rules out signal-to-noise ratios in physical units and the per-feature suitability metrics of prognostics [21, 22]. And it must connect to the

quantities a monitoring specification is written in: the smallest fault that shall be detected, the false-alarm rate tolerated and the sensors it costs.

This paper proposes such a measure, the signal-to-drift ratio (SDR): the relative change of an observable feature between a healthy reference interval and a faulted interval, divided by a quantile of the relative changes the same feature exhibits between adjacent healthy intervals when nothing is wrong. Statistically it is the known-onset, fixed-sample mean-shift test of change detection [23] expressed as a standardised effect size; an SDR of one marks the point at which a fault of a given extent becomes distinguishable from the ordinary non-stationarity of the healthy machine at a chosen per-trial false-alarm rate. Because the reference interval of every trial, faulted or not, contributes to the null, the SDR is calibrated on each machine's own healthy behaviour; dedicated healthy trials are needed only for its reliability screen. From the SDR follow the quantities a specification can be written on: the *minimum detectable extent* (MDE) of a fault class for a given machine type and sensor suite, the *detectable share* (DS) of trials above a required extent in which the fault is detected, and the *signature location*, the measurement group in which the fault appears. A decision table over the required extent and the false-alarm level turns them into the selection of the cheapest adequate sensor suite, and a transfer test establishes under which feature-referencing rule a detector built on one machine type serves another.

The question is made answerable empirically by a recent series of open benchmark datasets [24–26] in which three types of generator—a PMSG, an SCIG and a variable-speed WFSG—were subjected to the *same* inter-turn and inter-winding short-circuits, inserted at the same tapped positions of the stator winding; the PMSG and SCIG share one bench, one converter, one controller and one protocol, so that everything but the machine and the control structure it requires is held fixed. The datasets were published for algorithm benchmarking and refrain from comparing the machines; here the comparison is the object.

This paper makes four contributions. The first is the SDR as a measure of fault detectability that requires no model and no trained detector, with its statistical interpretation, its null pooled from the reference intervals of every trial, a binomial reliability screen with its empirical false-alarm counts, a calibrated oracle for multi-feature detection, and the three derived quantities MDE, DS and signature location, evaluated on the physically distinct fault sizes of a test matrix rather than on its nominal extents (Section 3). The second is a six-step procedure that turns these into a selection: a deci-

sion table that classifies every sensor suite of every machine type as meeting, failing or uncertain against a requirement on the extent, the detection power and the false-alarm level, with bootstrap confidence that carries the feature selection and the null, and a transfer test over feature-referencing rules for a detector that must serve several machine types. The third is a quantitative comparison, the first known to the author, of winding-fault detectability across three wind-turbine generator types under the same tapped short-circuits, with four findings: the design-level extent of a fault and its physical severity are different axes and must both be carried, and the faulted phase rides on the nominal extent axis unless sequence quantities are taken as phasor residuals; the measurement group in which a fault becomes visible depends on the machine–drive combination, and the detectability difference between the two machines on one bench comes from a quieter null in the stator current and from both signature and null in the controller quantities; the cheapest adequate sensor suite depends on the machine type *and* on the requirement; and a learned detector transfers across the machines once its features are referenced to healthy behaviour of the target machine, best to each trial's own immediately preceding healthy interval. The fourth is a sensitivity and uncertainty analysis of the tuning choices (null quantile, window length, screen), with trial-level and tap-pair-level bootstrap intervals, and open code that reproduces every result from the public data.

The paper is organised as follows: Section 2 positions the SDR among the detectability measures of FDI and health monitoring and summarises what is known about winding-fault signatures in converter-fed machines; Section 3 defines the SDR, the derived quantities and the six-step procedure; Section 4 describes the three machines, the fault protocol, the sensor suites and the computation; Section 5 reports the results; Section 6 draws out the implications for the condition monitoring of wind-turbine generators and the limits of what one machine per type supports; and Section 7 concludes. Appendix A gives feature definitions, computation details and four additional tables.

2. Background and related work

2.1. Detectability measures in fault detection and health monitoring

Fault detection and isolation has a mature theory of diagnosability for systems described by models. Structural analysis determines which faults can be detected and isolated from which sensor sets without numerical parameter values [16, 17], sensor placement selects the cheapest sensor set that

makes a required fault set detectable [18], and the approach has been applied to electric-vehicle drives with a diagnosability index that ranks candidate residual generators [27]; testability analysis in the test-engineering tradition records the same information as a dependency matrix of tests against faults [28, 29]. Basseville [15] gives statistical definitions of detectability and isolability for parametric models as distances between the distributions of a residual under the healthy and the faulted hypothesis. All of these presuppose a model whose structure is known and trusted [30, 31]. Data-driven diagnosis [19] replaces the model by data from the very machine being monitored, and its transfer across machines is an active problem [20]. The prognostics and health management (PHM) community grades candidate features by monotonicity, trendability and prognosability [21], ranks them by separability measures such as the Fisher criterion or the per-feature area under the receiver operating characteristic (ROC) curve [32], evaluates prognostic algorithms by dedicated metrics [22] and has reviewed the sensor systems that feed them [33]. None of these yields, for a fault of known extent in a built or simulated machine, a number that states how far above the healthy non-stationarity of each candidate measurement the fault stands at a stated false-alarm rate, comparable across measurements in different units and across machines.

Table 1 places the SDR among these methods. It is a per-feature separability measure of the same family as the sensitivity index d' of signal detection, the fault-to-noise ratio of model-based FDI [15] and the per-feature AUC; it requires neither a model structure nor a trained detector, only trials with a known fault extent and observable features. What distinguishes it is the null it is calibrated on, the drift of each feature between adjacent healthy intervals of every trial, faulted or not, which needs healthy-only recordings only for the reliability screen, fixes the false-alarm rate of the simplest detector and makes the magnitude comparable across features, operating points and machines; and what is new is that calibration, the reliability screen, the limit-of-detection quantities derived from the statistic, and their use to compare machine types and sensor suites and to select among them (Section 3.2 states the relation to the mean-shift test of Basseville and Nikiforov [23]). It deliberately stops at detectability: isolation, severity estimation and prognosis presuppose it.

2.2. Stator winding faults in converter-fed machines

Inter-turn short-circuits in stator windings are among the most studied incipient faults in electrical machines [1, 2, 34]. Motor-current signature

Table 1: Positioning of the signal-to-drift ratio among detectability and feature-assessment methods

Method	Input	Output	Needs model or detector	Compares machines and sensor suites
Structural diagnosability, sensor placement [16–18]	model structure	detectable and isolable fault sets per sensor set	model structure	per model, no magnitude
Statistical detectability [15]	parametric model, residual	distance between hypotheses	model	per model
Testability analysis [28, 29]	test–fault dependency matrix	detectable and isolable fault sets, test sequences	dependency model	no
PHM feature suitability [21]	run-to-failure data	feature scores	no	no
Per-feature separability (d', Fisher score, AUC [32])	labelled trials	score per feature	no	across features of one machine; across machines only with a common noise scale
Learned detectors [19]	labelled data of the machine	detector and its ROC	trained detector	only through transfer [20]
This paper	trials with known fault extent (measured or simulated)	MDE, detectable share and signature location per machine type and sensor suite; decision table; transfer rule	neither	yes, scale-free and self-calibrated

MDE: minimum detectable extent; PHM: prognostics and health management; ROC: receiver operating characteristic.

analysis [5], negative-sequence current and impedance methods [6, 7, 35], the extended Park's vector approach [36], turn-fault models [37] and the review literature [8, 9, 38, 39] have established the physical signatures: an asymmetry of the winding produces negative-sequence components in the terminal quantities and a component at twice the fundamental frequency in the synchronous reference frame, and because the fault-induced negative-sequence phasor adds to the inherent asymmetry of the machine, the detection quantity is the change of that phasor rather than of its magnitude [6, 35]. Under closed-loop current control the current regulator acts against the negative-sequence current it can see, so the asymmetry migrates from the currents

into the voltage commands and the controller states, with the degree of migration set by the current-loop bandwidth relative to twice the electrical frequency [10, 11]; the closed-loop negative-sequence impedance that results is itself a sensitive detection quantity in current-regulated drives [40]. In a wound-field synchronous machine the stator negative sequence additionally induces a component at twice the electrical frequency in the field current [41]; in permanent-magnet machines with an accessible neutral the zero-sequence voltage is a sensitive indicator, also under field-oriented control [42, 43]; and the stray flux outside the frame is the other established non-intrusive route [39]. These facts are well known to drive specialists but have not been turned into a measure that can be applied across candidate machine types. Machine-learning detectors are typically developed for one machine [44, 45], and open datasets of winding faults exist for one machine type at a time [46]; the transfer experiment in this paper is the cross-machine counterpart of such detectors. Finite-element turn-fault models [47] and digital twins [48] are the natural trial generators for an instantiation of the method before hardware exists.

2.3. Generator type as a design decision

The generator concept of a large wind turbine—direct-drive or geared; permanent-magnet, induction, wound-field or doubly-fed; full or partial power conversion—is selected early on cost, mass, efficiency and grid-compliance grounds [12, 13]. Reliability enters the trade through failure-rate statistics of subassemblies [3, 4], and condition monitoring is added afterwards from a catalogue of commercial systems built around vibration, oil and electrical signals [14, 49], not through the observability of the failures in each concept. The design community's concept-stage failure reasoning—function–failure mapping and functional failure identification and propagation [50–52]—and the detection rating of a design failure mode and effects analysis record which failures a concept can suffer and an expert's judgement of whether they would be detected, not a measured magnitude. Because monitoring designs increasingly serve a family of variants [53], whether a monitoring design transfers across machine types is a family-level question, which the transfer step of the method addresses.

3. Method

3.1. Definitions and terms

Let $\mathcal{A}$ be a set of candidate *alternatives*. An alternative $a \in \mathcal{A}$ is a machine type together with the control structure it implies—for a generator, the machine topology and the converter control that the topology requires. Let $\mathcal{F}$ be a set of failure-mode classes; each class $f \in \mathcal{F}$ is parameterised by an *extent* $e \in E_f$ expressed at the design level, so that the same (f, e) can be realised in every alternative: for a winding fault the fraction of a winding branch that is short-circuited, for a bearing a defect size, for a structure a crack length. The design-level extent does not fix the physical severity of the event (Section 5.1), so the framing also names a *physical severity axis* on which the alternatives can be compared. Let $\mathcal{S}$ be a set of candidate *sensor suites*; a suite $s \in \mathcal{S}$ is a set of measurements, including quantities internal to the drive controller, with a cost $c(s)$, and $\Phi(s)$ is the set of features computable from those measurements alone. Let $\mathcal{O}$ be the operating envelope, discretised into operating points o.

A *trial* is a record of the system in alternative a at operating point o containing a reference interval R in which the system is healthy and, immediately after it, a faulted interval F in which failure mode (f, e) is present. Trials may come from a physical test (test-stage instantiation) or from a validated simulation (design-stage instantiation, Section 3.5). These terms—alternative, trial, sensor suite, extent (design level) and severity (physical)—are used throughout.

3.2. The signal-to-drift ratio

The reference interval is split into two adjacent halves R_1, R_2, with R_2 adjacent to F. For a feature $\varphi \in \Phi(s)$ and a trial, let $\bar{\varphi}_{R_1}, \bar{\varphi}_{R_2}, \bar{\varphi}_F$ be the feature's mean values over the intervals, and define the relative change under fault, referenced to the half adjacent to the fault, and the relative change between the two healthy halves,

$$\delta_\varphi = \frac{\bar{\varphi}_F - \bar{\varphi}_{R_2}}{|\bar{\varphi}_{R_2}|}, \qquad \delta^0_\varphi = \frac{\bar{\varphi}_{R_2} - \bar{\varphi}_{R_1}}{|\bar{\varphi}_{R_1}|}. \tag{1}$$

δ^0_φ measures how much the feature moves when nothing is wrong: sensor noise, drift of the operating point, controller activity, thermal effects. Because every trial has a healthy reference interval, it is available from every trial of alternative a, faulted or not, and its absolute value pooled over the

N_a trials forms an empirical null distribution of ordinary non-stationarity for that alternative and feature. The signal-to-drift ratio is

$$\mathrm{SDR}_\varphi = \frac{|\delta_\varphi|}{q_\alpha\big(|\delta^0_\varphi|\big)}, \tag{2}$$

where q_α is the α-quantile of the pooled null; with $\alpha = 0.95$, $\mathrm{SDR}_\varphi \geq 1$ means that the change under fault exceeds what the healthy system produces on its own in 95 % of comparable interval pairs.

What the SDR is statistically. The SDR is a mean-shift statistic with a known change time, studentised by an empirical quantile of its own drift null rather than by a within-interval standard deviation. For a zero-mean Gaussian null $q_{0.95}(|\delta^0|) = 1.96\,\sigma_{\delta^0}$, and $\mathrm{SDR} \geq 1$ is then a two-sided 5 % test on the shift of interval means, the fixed-sample, known-onset special case of the generalised likelihood ratio of Basseville and Nikiforov [23]; on these data the quantile and the Gaussian scale agree to within 12 % for the scalar features ($q_{0.95}/1.96\sigma$ of 0.95 to 1.12, Table A.4), but the quantile is kept because the null has an excess kurtosis of up to 7 and at $\alpha = 0.99$ is set by the largest two or three of 225 values. In the language of signal detection the SDR is the sensitivity index d' of the interval-mean shift against the drift distribution; in that of model-based FDI it is the fault-to-noise ratio of Basseville [15] with the noise scale taken from the healthy records instead of from a model; the detectable share defined below is an empirical power, and a share at a given α is one point of a receiver operating characteristic. The quantile is an estimate: its 95 % sampling interval spans a factor 1.3 to 1.5 for the key features (Table A.4), and every interval reported in this paper resamples it together with the numerator. Three assumptions are built in: the onset is known, the whole faulted interval is averaged, and the healthy reference lies immediately before the fault; the last makes the null one of sub-second non-stationarity, and Section 5.8 quantifies how much wider the null of a reference established at commissioning is. The SDR characterises the simplest detector on the feature—a threshold on the relative change of interval means at a known onset—and the design quantities derived from it are statements about that detector, an abrupt-onset change detector at a stationary operating point; a detector that combines features or accumulates evidence over time can detect below $\mathrm{SDR} = 1$, which is why the calibrated oracle and the transfer experiment of step 6 are part of the method.

Four properties make the SDR suitable as a measure for comparing alternatives and suites: it is *scale-free*, so a current ratio, a voltage harmonic

and a controller output are compared on one axis; *calibrated* against each alternative's own non-stationarity, so a noisier alternative is not credited with detectability it does not have, at the price that numerator and denominator must be reported separately when alternatives are compared (Section 5.2), because a quieter null raises the SDR as much as a stronger signature does; *per trial*, so its distribution over trials carries the dependence on operating point and extent; and *detector-free*, so it can be computed before any algorithm is chosen. The relative form presumes a non-zero healthy level; for features whose healthy value is a noise floor an absolute-change variant with the same structure is used, and Section 5.8 shows that the two rank the features alike.

Safeguards. (i) *Reliability screen.* On healthy trials—trials in which the "faulted" interval contains no fault, only the switching operation that would insert one—a feature should exceed SDR = 1 at the nominal rate $1 - \alpha$. A feature whose count of such false alarms in the n_h healthy trials is incompatible with that rate (one-sided binomial test at the 5 % level; three or more of nine at $\alpha = 0.95$) is excluded from the design quantities for that alternative, as are features whose null quantile exceeds one (healthy relative change above 100 %). With nine healthy trials the screen has little power—a feature whose true false-alarm rate is 20 % passes it with probability 0.74—so the empirical false-alarm count of every fixed feature is reported beside its design quantities. (ii) *Fixed, nested and oracle feature.* A suite is evaluated by one *fixed* feature—chosen once among the suite's screened features as the one with the smallest minimum detectable extent, ties broken by the detectable share and then by the median SDR, so that a suite is never credited with less than its subsets—or by the *oracle* feature, the best feature of each trial. The fixed variant is what a threshold detector achieves; the oracle variant bounds what a detector that combines the suite's k features can achieve, and because a maximum over k features exceeds the single-feature threshold at a rate of up to $1 - \alpha^k$, it is reported at a calibrated threshold, the α-quantile of $\max_\varphi |\delta^0_\varphi| / q_\alpha(|\delta^0_\varphi|)$ over all records of the alternative, which holds its false-alarm rate at $1 - \alpha$. Because the fixed feature is chosen on the trials it is evaluated on, a *nested* variant chooses it on all but one tap pair and evaluates it on the held-out pair. (iii) *Uncertainty.* Every design quantity carries a bootstrap interval: over tap pairs (the physical fault locations) for statements about a class and over operating points within an extent level for the minimum detectable extent, with the null quantile re-estimated from a cluster resample of all records in every resample, so that the intervals carry

the sampling variability of the denominator as well as of the numerator.

3.3. Derived quantities

From the SDR distribution over trials the following are defined for each alternative a, suite s and class f:

- the *diagnosability map* $D_a(s, f, e, o)$: the SDR of the suite's fixed feature at extent e and operating point o;
- the *minimum detectable extent* $\mathrm{MDE}_a(s, f)$: the smallest tested extent level e^* such that $\mathrm{median}_o\, D_a(s, f, e, o) \geq 1$ for every tested $e \geq e^*$ (limit-of-detection convention), reported as the bracket $(e^-, e^*]$ in which e^- is the largest tested level below e^*, because on a tested grid the MDE is known only to lie between the last undetected and the first detected level; the levels are the physically distinct fault sizes of the test matrix (Section 4.2), not the nominal extents;
- the *detectable share* $\mathrm{DS}_a(s, f \mid e \geq e_{\mathrm{req}})$: the fraction of faulted trials with $e \geq e_{\mathrm{req}}$ in which SDR ≥ 1, written for a required extent so that it does not depend on how the test matrix samples the extent axis;
- the *signature location*: the measurement group—stator currents, terminal voltages, controller internals, mechanical quantities, excitation—whose best screened feature attains the highest SDR in the largest share of trials.

The *alternative-dependent diagnosability* of a set of alternatives is then the map that assigns to every triple (a, s, f) the triple $(\mathrm{MDE}_a, \mathrm{DS}_a, \text{location})$. For a requirement $(f, e_{\mathrm{req}}, \alpha)$ the *adequate* suites of alternative a are those with $\mathrm{MDE}_a(s, f) \leq e_{\mathrm{req}}$, and diagnosability is alternative-dependent for that requirement when the cheapest adequate suite differs between alternatives by more than the bootstrap uncertainty of the MDE. MDE and DS are the quantities against which requirements are written ("inter-turn faults of extent $\geq e_{\mathrm{req}}$ shall be detectable over the operating envelope at a per-trial false-alarm rate $\leq 1 - \alpha$"); the signature location tells the designer *why* an alternative behaves as it does.

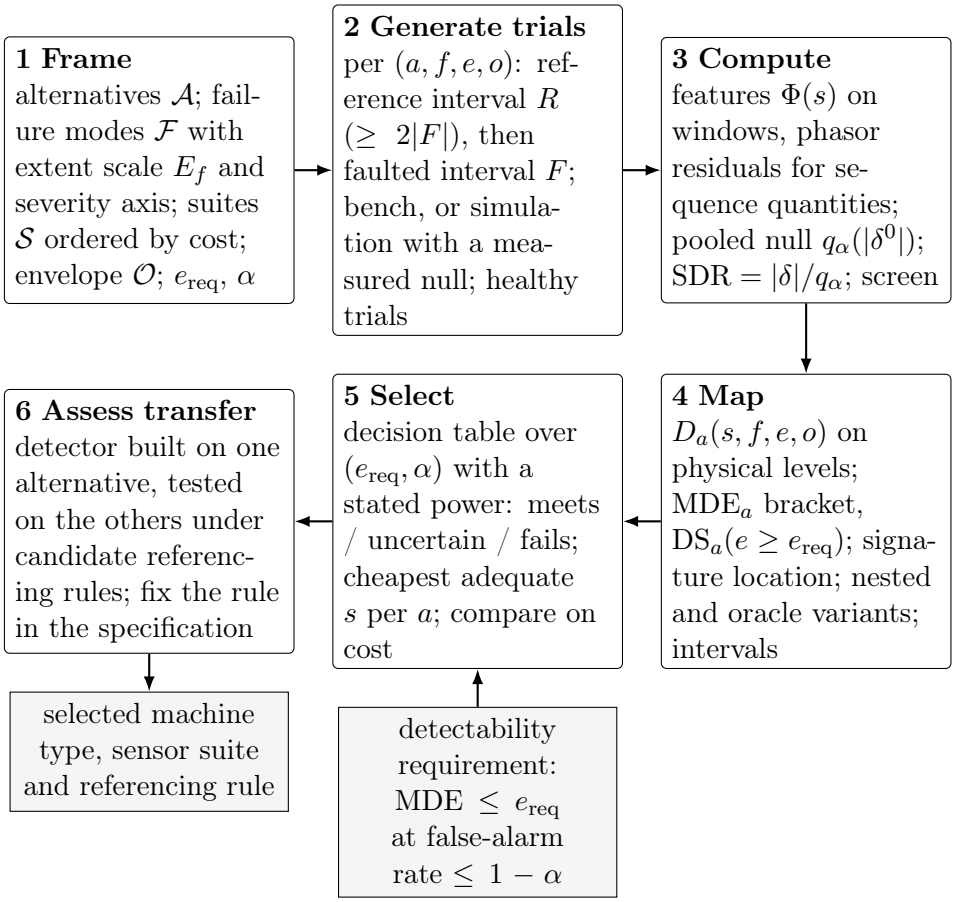


Figure 1: The six-step procedure; steps 3–4 are fixed by Eqs. (1)–(2), steps 1, 2, 5 and 6 follow the default rules of Section 3.4

3.4. Procedure

The procedure (Fig. 1) has six steps. Steps 3–4 are mechanical and fully specified by Eqs. (1)–(2); steps 1, 2, 5 and 6 are design decisions, for which the method supplies default rules.

1. **Frame.** Enumerate $\mathcal{A}$, the classes $\mathcal{F}$ with an extent scale E_f realisable in every alternative and a physical severity axis, the suites $\mathcal{S}$ and the envelope $\mathcal{O}$. Set e_{req} as the smallest extent whose growth to an unacceptable consequence is faster than the inspection interval, and $1 - \alpha$ as the per-trial false-alarm rate the operator tolerates over that interval. Order the suites by cost: an ordinal rank by added hardware

and integration by default, a monetary cost including installation and maintenance when available.

2. **Generate trials.** For each (a, f, e, o) obtain trials with a healthy reference interval of at least twice the faulted duration, immediately followed by the faulted interval; test at least the extent levels that bracket e_{req} and the corners of the envelope; include enough healthy trials with the same timing for the screen (nine allow the binomial rule above). At the test stage these are prototype or benchmark experiments; at the design stage the faulted numerator is simulated and the null is supplied as in Section 3.5.
3. **Compute.** Extract $\Phi(s)$ for all suites on sliding windows within the intervals, form the pooled null per alternative and feature, compute the SDR per trial, apply the screen.
4. **Map.** Build $D_a(s, f, e, o)$ and summarise it into MDE_a, DS_a and the signature location for the fixed, nested and oracle variants, with bootstrap intervals.
5. **Select.** Build the decision table: for each alternative, class and suite and for the candidate values of e_{req} and α, classify the suite as *meets* when the bootstrap probability of the joint event $\{\text{MDE} \leq e_{\text{req}}$ and $\text{DS}(e \geq e_{\text{req}}) \geq \text{DS}_{\text{req}}\}$ is at least 0.95, *fails* when it is at most 0.05, and *uncertain* otherwise; the MDE alone is a 50 %-power quantity (the median trial of a level), so the required share DS_{req} states the power the requirement demands (0.9 here). In every resample the fixed feature is re-selected and the null re-estimated, so the probability carries the selection step, and because a suite may always use the features of its subsets, its verdict is at least that of its subsets; the table is printed with the probabilities, not only the verdicts. The candidate values of e_{req} are the tested extent levels, because between two levels the MDE is known only as a bracket, and a requirement outside the range of extents an alternative was tested at is reported as not testable rather than as failed. The adequate suite of an alternative is the cheapest that meets the requirement for *every* class; an uncertain verdict is resolved by more trials around e_{req} or by specifying the next suite as a fallback. The alternatives are then compared on the cost of their adequate suites together with the other selection criteria; the choice of machine type is affected only when, for some alternative, no affordable suite meets the requirement, otherwise the method changes the monitoring specification and says so.

6. **Assess transfer.** If the suite and its detector must serve a family of alternatives, test whether a detector built on one alternative's trials detects the others' faults under the candidate feature-referencing rules (physical units; scaled on the target's healthy trials; referenced to each trial's own preceding healthy interval). The rule that transfers becomes part of the monitoring specification.

3.5. Design-stage instantiation

The demonstration in this paper is a test-stage instantiation on built machines. A design-stage instantiation, before hardware exists, must supply the two ingredients of the SDR separately: the numerator, the change of a feature under a fault of extent e, is what a finite-element or circuit model of each alternative with the fault injected provides [47]; the denominator, the healthy non-stationarity of sensor noise, controller activity, thermal drift and operating-point wander, is a property of hardware and environment that such a model does not reproduce, so a simulated SDR that supplied its own null would be optimistic by an unknown factor. The null must come from measured healthy records of hardware of the same class (a predecessor product, a prototype or a benchmark such as this one), on the testable assumption that it depends on the sensing and control chain more than on the machine (Table 4 compares the nulls of the two machines on one bench), and the simulated numerator is validated against measured SDRs on a few physical trials.

3.6. Scope

The method applies to failure-mode classes that can be seeded at a known extent, at a stationary operating point, with a healthy interval immediately before the fault, in every alternative; the extent scale must be realisable in all of them, and the physical severity axis is declared beside it. It evaluates detectability by a threshold on interval means, not the time to detection, and stops at detection: isolation, severity estimation and remaining-life prediction presuppose detectability and can be built on the same trials. The false-alarm level $1 - \alpha$ is a per-decision rate for one faulted interval; at one decision per interval length (0.36 s here) $\alpha = 0.95$ corresponds to a false alarm every 7 s of operation, so the values of α used in a test-stage instantiation characterise the detectability of the bench, and a field monitor needs an accumulation rule over many decisions that brings the per-alarm rate to the order of one per month; the MDEs reported with $\alpha = 0.95$ are therefore bench quantities,

to be read as limits of detection, not as field thresholds. It does not model the cost of missed detection, which enters through e_{req} and α. Run-to-failure data, in which the fault grows gradually and no healthy interval precedes a known onset, do not fit the trial structure and would need a different null.

4. Demonstration on three wind-turbine generators

4.1. Alternatives and data

The three alternatives are laboratory-scale generators of the same power class on instrumented test benches at the University of São Paulo, published as open benchmark datasets (Table 2, Fig. 2). The PMSG [24] and the SCIG [25] are four-pole, 230 V, double-star machines from the same manufacturer, rated 2.5 kVA and 2.5 kW, installed on the *same* bench: the same induction-motor prime mover and frequency converter, the same rapid-control-prototyping platform and field-oriented control (FOC; rotor angle from an encoder for the PMSG, from a flux observer for the SCIG), the same sensors, acquisition (20 kHz, 3 s records), fault-insertion hardware and test script. The variable-speed WFSG [26] is a 2 kVA four-salient-pole machine on an earlier bench of the same group [54] with a three-level converter, FOC with encoder angle, 4 kHz acquisition and 1.3 s records. The PMSG–SCIG pair is therefore a controlled comparison in which everything but the machine and the control structure it requires is held fixed. The WFSG is a third alternative with the same tapped short-circuits on a different bench: its fault resistances, fault durations, converter, sampling rate, record length and voltage sensing all differ (Table 2), its variable-speed dataset holds no healthy records, so the reliability screen and the nuisance-transient negatives are unavailable for it, and it is analysed as a third case at matched extents with those differences kept explicit rather than as a replicate of the controlled pair. The current-loop gains and bandwidths, the switching frequencies and the structure of the SCIG's flux observer are not stated in the data papers. Terminal voltages are measured by each bench's voltage transducers, whose bandwidth the data papers do not document, so the terminal-voltage features carry the sensing chain of their bench as well as the machine.

4.2. Failure-mode class and extent scale

All three machines were built with 24 tapped derivation points along their stator windings at the same relative positions, so that a short-circuit between two taps realises a fault of known extent; the tap positions are identical in the

Table 2: The three alternatives; the PMSG and SCIG share one bench and one protocol, the WFSG bench differs where indicated

	PMSG	SCIG	WFSG (variable speed)
Rating, poles	2.5 kVA, 4	2.5 kW, 4	2 kVA, 4 salient
Excitation	permanent magnets	induction (magnetising current)	wound field, static exciter
Converter and control	2L, FOC with encoder	2L, FOC with flux observer	3L-NPC, FOC with encoder
Acquisition	20 kHz, 3 s	20 kHz, 3 s	4 kHz, 1.3 s
Reference / faulted interval	0.68 s / 0.36 s	0.68 s / 0.36 s	0.49 s / 0.06–0.28 s
Fault impedance	2.6 Ω	2.6 Ω	1, 2.83, 5.66, 11.32 Ω
Operating points	3 speeds × 3 torques	same	716–1910 rpm, 4 torque levels (sparse)
Healthy trials	9	9	none
Fault trials	216	216	438 (198 at 2.83 Ω)

Ratings as given by the manufacturer (kVA for the synchronous machines, kW for the induction machine). 2L: two-level converter; 3L-NPC: three-level neutral-point-clamped converter; FOC: field-oriented control. Current-loop gains and bandwidths, switching frequencies and the flux-observer structure are not given in the data papers.

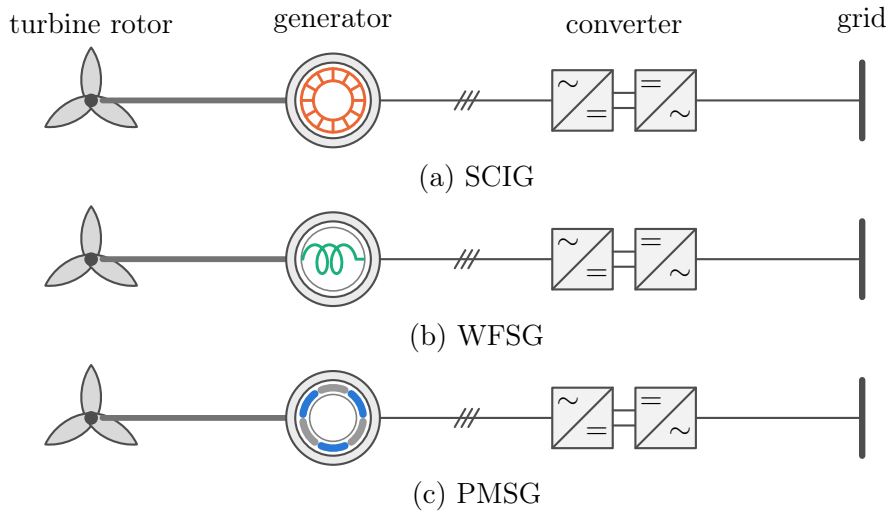
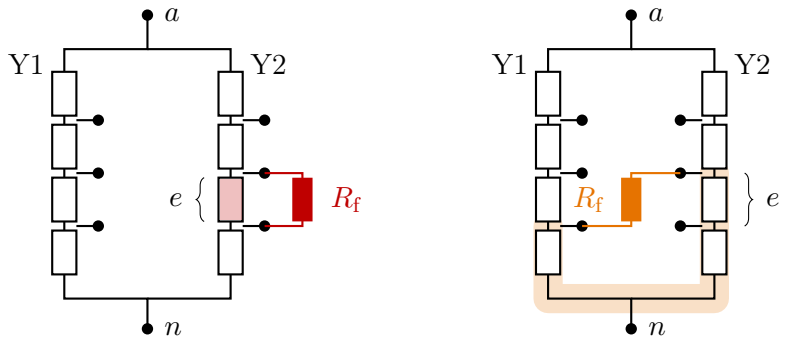


Figure 2: The three alternatives as generator systems: turbine rotor, generator with its rotor excitation (cage, field winding or magnets), converter and grid

three data papers, which is what makes the extent scale realisable in every alternative. Two classes are used (Fig. 3): inter-turn short-circuits (both taps on the same branch) and inter-winding short-circuits (taps on the two parallel

branches of a phase). The extent e is the fraction of a branch between the two taps; it ranges from 2.7 % to 12.1 % for the twelve inter-turn cases and from 2.3 % to 40.2 % for the twelve inter-winding cases. The nominal extents, however, overstate the resolution of the test matrix. The fault current of a resistor-limited short-circuit at a given operating point is set by the voltage induced in the shorted turns, so tap pairs with equal fault currents short the same number of turns and are the same physical fault: on the PMSG the four inter-turn pairs at 2.7–3.8 % all carry 1.41–1.44 A, the four at 7.4–7.7 % 3.74–3.77 A and the four at 11.6–12.1 % 6.0 A (Section 5.1), and the SCIG currents cluster identically. The limit-of-detection quantities are therefore evaluated on *physical extent levels*: tap pairs of one class whose median fault currents, normalised per machine and averaged over the two machines that share the taps, lie within 7 % of the first pair of the level. This gives three inter-turn levels (2.7–3.8, 7.4–7.7 and 11.6–12.1 % of a branch) and six inter-winding levels (2.3–3.2, 9.7–10.6, 14.3–14.8, 21.7–22.2, 33.7–37.5 and 40.2 %), each labelled by the range of nominal extents it spans; the nominal extent remains the abscissa of the figures. The tap layout has a second property that matters for sequence features: the four smallest inter-turn pairs all lie in phase A, the 7.4–7.7 % pairs in phases A and B and the largest in phases A and C, so on the nominal extent axis the faulted phase is confounded with the extent (Section 5.3). Faults were closed through a contactor and a series resistance after the machine had reached steady state: 2.6 Ω for 0.4 s on the PMSG and SCIG; on the WFSG the experimenters raised the resistance with extent (1 Ω for the inter-turn cases below 11 %, 2.83 Ω for the remaining inter-turn cases and the small inter-winding cases, 5.66 and 11.32 Ω for the large inter-winding cases), so that on that bench extent and impedance are confounded. The fault current is measured on all benches; it is used only as ground truth for the physical severity axis, never as a feature. Because the series resistance is of the order of the winding impedance of these small machines, the fault current is essentially the voltage induced in the shorted turns divided by the resistance: a *resistor-limited* severity, unlike a bolted fault, which the leakage reactance of the shorted turns limits (Section 5.1).

The PMSG and SCIG matrix is complete: 24 fault cases × 3 speeds (1200, 1500, 1800 rpm) × 3 torques (5.2, 6.4, 8.0 N m), the torque–speed pairs scaled from the operating region of the International Energy Agency (IEA) 15 MW reference turbine [55] to the laboratory speed range (the electrical frequency is not scaled), plus the healthy condition at each operating point; the SCIG generates with slip, at 59.6 Hz at 1800 rpm. The WFSG matrix covers the



(a) Inter-turn short-circuit (b) Inter-winding short-circuit

Figure 3: Inter-turn (**a**) and inter-winding (**b**) short-circuits through R_f between taps of one phase of the double-star stator; e is the fraction of a branch between the taps

same 24 tap pairs at six speeds (716–1910 rpm) and four per-unit torque levels, sparsely and with two to six repetitions, so its operating points are not matched to the other two. The WFSG benchmark also contains phase-to-phase faults, a class absent from the PMSG/SCIG protocol; they are excluded from every analysis. The analyses use the WFSG's 438 inter-turn and inter-winding trials at all impedances for the null, the feature ranking and the detectable shares, with the impedance as a covariate, and the 198 trials at 2.83 Ω—the impedance closest to the PMSG/SCIG value, covering four inter-turn tap pairs of 11.6–12.1 % and four inter-winding pairs of 2.3–10.6 %—for the minimum detectable extent and the decision table.

4.3. Sensor suites and features

Five sensor suites are considered, ordered by the data access, hardware and integration they add to a bare generator–converter set (Table 3). *Drive-internal* uses only quantities the converter controller already has; *3 CT* (three current transducers) uses the three phase currents—from dedicated transducers or from the converter's own current sensors routed to a monitoring system—with angle-free features, so that for a converter-fed machine its cost is data access and bandwidth rather than hardware; *3 CT + 3 VT* adds three terminal voltage transducers (VT); *3 CT + 3 VT + drive* adds controller access, which also allows the measured terminal voltages to be transformed into the synchronous frame; and *+ torque transducer* adds a shaft torque sensor. For the WFSG the field current is an additional drive-internal observable (the *excitation* suite), available because its exciter is static [41]. Each feature is

Table 3: Sensor suites and the features they can compute

Suite	Added access / hardware	Features
Drive-internal	controller data	I_d, I_q: std, $1f_e$, $2f_e$; $V_d^{\mathrm{cmd}}, V_q^{\mathrm{cmd}}$: $2f_e$; PI outputs d, q: std, $2f_e$; duty-cycle negative sequence D_2/D_1; T_e: std, $1f_e$, $2f_e$; DC-link std; speed std
3 CT	three phase currents, no angle	I_2/I_1, RMS unbalance, harmonics 3, 5, 7 of I_a relative to the fundamental, THD
3 CT + 3 VT	+ 3 voltage transducers	3 CT features + V_2/V_1
3 CT + 3 VT + drive	+ controller data	all of the above + measured V_d, V_q: $2f_e$
+ torque transducer	+ shaft torque sensor	+ T_m: std, $1f_e$, $2f_e$
Excitation (WFSG)	field-current measurement	I_f: std, $2f_e$

f_e: electrical frequency; $1f_e$, $2f_e$: amplitude of the component at once and twice the electrical frequency; I_1, I_2 (V_1, V_2): positive- and negative-sequence magnitudes at f_e; std: standard deviation; THD: total harmonic distortion; PI: proportional–integral.

assigned to the cheapest suite whose measurements suffice to compute it. Sequence quantities enter in two forms. The magnitude ratios I_2/I_1, V_2/V_1 and D_2/D_1 are the established negative-sequence indicators [6, 7], and the angle-free I_2/I_1 is the quantity behind the extended Park's vector approach [36]. Because the fault-induced negative-sequence phasor adds to the machine's inherent asymmetry as a complex number, the change of a magnitude ratio under fault depends on the angle between the two, which rotates with the faulted phase, and the residual of the complex phasor against its healthy value is the established remedy [6, 35]; the phasor residuals $|\Delta\bar{I}_2|/I_1$, $|\Delta\bar{V}_2|/V_1$ and $|\Delta\bar{D}_2|/D_1$ therefore take the negative-sequence component referenced to the positive-sequence angle and the magnitude of its change between the intervals (Appendix A.1), which isolates the fault-induced part whatever the faulted phase (Section 5.3). The suites are those the datasets can populate (zero-sequence voltage, stray flux, winding temperature, vibration and partial discharge are not recorded, and the angle-free Park's-vector features were not computed), so the suite conclusions hold among the suites and features considered; the zero-sequence current is not used because the star points are isolated.

Features are computed on sliding windows of n_c electrical cycles with a hop of one cycle; the electrical frequency and the phase rotation are estimated

per window from the current space vector, so that the external suites need no angle sensor. Appendix A.1 defines every feature and states which WFSG channels are measured and which commanded.

4.4. Computation

For the PMSG and SCIG the reference interval is the 0.68 s before the contactor command, split into R_1 (0.32 s) and R_2 (0.36 s), and the faulted interval runs from 30 ms after the command to 10 ms before its release; for the WFSG the fault interval is located from the measured fault current and the reference interval (0.49 s) is split at 0.30 s. Appendix A.2 lists all interval boundaries, thresholds, seeds and model settings. The main PMSG–SCIG comparison uses $n_c = 5$; the three-alternative comparison uses $n_c = 3$ for all three machines because the WFSG fault intervals are short. The sensitivity of every conclusion to $n_c \in \{3, 5, 8\}$ and to $\alpha \in \{0.90, 0.95, 0.99\}$ is reported, with the screen applied at each setting. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 WFSG records, every record contributing a healthy reference interval. Uncertainty is quantified by bootstrap with 2000 resamples: the median SDR and the detectable share of a class are resampled over the twelve tap pairs, the same pairs for both machines so that the SCIG/PMSG ratio is matched; the minimum detectable extent is resampled over the operating-point trials within each extent level; and in every resample the null quantile is re-estimated from a cluster resample of the machine's records, so that the intervals carry the sampling variability of the denominator as well (Appendix A.2). The decision table uses 1000 resamples per suite, alternative, class and α, with e_{req} at the tested levels (2.7–3.8, 7.4–7.7 and 11.6–12.1 % of a branch for inter-turn faults and the six inter-winding levels) and a required detectable share of 0.9 above the requirement.

Step 6 of the procedure is instantiated as a window-level change detector: a gradient-boosted decision-tree (GBDT) classifier [56] and, for the PMSG–SCIG pair, a logistic regression, trained to separate faulted windows from healthy ones. Within an alternative the detector is evaluated by five-fold cross-validation grouped by tap pair (healthy trials grouped by speed), so that the tested fault locations are never seen in training; across alternatives it is trained on all trials of one machine and tested on all trials of another. Four feature-referencing rules are compared: physical units; features z-scored on all healthy trials of the target machine (a commissioning baseline); z-scored

on the healthy trial recorded at the same operating point (a per-operating-point baseline); and the absolute z-score of each window against the *first half* of its own trial's reference interval, the operational form of the SDR's referencing assumption. Negatives are the reference and recovery windows of all trials plus the fault-interval windows of the healthy trials, the only negatives that contain the contactor operation without a fault. Because the windows used to normalise a trial are trivially separable under the fourth rule, every metric is reported on the out-of-reference windows, with 95 % intervals from a cluster bootstrap over the tested machine's trials, and the false-alarm rate of the threshold that gives 1 % false positives on all negatives is reported on the healthy trials' fault-time windows; the true-positive rate at 1 % false-positive rate is a point of the receiver operating characteristic, not an operating threshold.

All computations are implemented in Python and released with the paper; the pipeline runs from the feature caches to every table and figure in under an hour on a four-core machine.

5. Results

5.1. The same design-level fault is not the same physical fault

For the same tap pair the SCIG's fault current is about 10 % lower than the PMSG's (medians 3.3 against 3.7 A for inter-turn and 7.7 against 8.5 A for inter-winding pairs), but the difference that matters for detection is relative to what the stator sensors see: the fault current is 1.45 times the pre-fault stator current (RMS to RMS, median over pairs) in the PMSG and 0.63 times in the SCIG for inter-turn faults, and 3.3 against 1.5 times for inter-winding faults, because the SCIG's stator current carries the magnetising component (5.3 against 2.6 A RMS before the fault). The WFSG trials at the matching impedance give 1.20 (inter-turn, 11.6–12.1 %) and 1.03 (inter-winding, 2.3–10.6 %), and 0.65 at the lower impedance used for its small inter-turn extents. Being resistor-limited (Section 4.2), these currents scale with volts per turn and speed; a bolted fault would reach several times rated current and—the first-order difference between the topologies—could be de-excited in the SCIG and the WFSG but not in the PMSG. A fault specified as "e % of a winding" therefore arrives at the sensors of the three alternatives as three different physical events, a property of the resistor-limited emulation as much as of the machines; Section 5.4 uses both axes. The physical severity axis, the RMS fault current relative to the RMS stator current, orders faults of equal

turns correctly because the MMF asymmetry of a turn fault scales with the fault current times the shorted fraction, so within a physical level the current alone ranks the trials; across levels it is a measured, not a general, severity scale, and the equal-current clusters of Section 4.2 are its direct consequence.

5.2. Where the signature appears, and why the SDR differs

The null of the SDR differs between the PMSG and the SCIG on every feature, and by how much depends on the form in which the feature is written (Table 4). In the relative form of Eq. (1) the SCIG's null is 2.4 times quieter than the PMSG's for the voltage residual, 5.2 times for the current residual and 45 times for the d-axis current ripple; in the units of the features themselves the factors are 1.1, 6.0 and 7.0. The two forms disagree where the healthy level of a feature differs between the machines: the SCIG's d-axis ripple has a healthy level five times the PMSG's (0.097 against 0.018 A), because its stator current carries the magnetising component, so a null that is 7 times quieter in amperes becomes 45 times quieter in relative terms, and a fault-induced change that is 1.9 times the PMSG's in amperes becomes 0.37 times it in relative terms. The decomposition of the SCIG/PMSG ratio of median SDR therefore reads differently for the three measurement groups. Through the terminal-voltage residual the two machines are equivalent: the ratio is 1.01 for inter-turn and 0.74 for inter-winding faults, and in volts numerator and null agree between the machines to within 20 %. Through the stator-current residual the SCIG's SDR is 3.7 (inter-turn) and 3.3 (inter-winding) times the PMSG's; the PMSG's current change is the larger one in both forms (SCIG/PMSG ratio 0.71 relative, 0.78 absolute), so here the SCIG wins entirely by a null five to six times quieter. Through the d-axis features the SCIG's SDR is 16 and 23 times the PMSG's, and in absolute units the SCIG has both the larger signature (1.9 and 3.9 times the PMSG's ripple in amperes) and the quieter null (7 times): the reading that the SCIG wins only by silence holds for the stator current and is, for the controller quantities, an artefact of the relative normalisation. The null is not a property of the transducers alone: the duty-cycle residual, a controller output, has a null 8.5 times larger on the PMSG in relative terms and 3.1 times in absolute terms, so the healthy non-stationarity belongs to the machine–drive combination.

Fig. 4 shows the median SDR of every feature for the three alternatives (the WFSG pooled over its fault impedances; features to the right of the dashed line at SDR = 1 are detectable in more than half of the trials). In

Table 4: SCIG/PMSG ratio of median SDR decomposed into numerator and denominator, five-cycle windows

Class	Feature	relative (SDR)			absolute units		
		Num.	Den.	SDR	Num.	Den.	SDR
IT	V_2/V_1	0.53	2.85	1.51	1.04	0.99	1.03
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	0.42	2.40	1.01	0.84	1.10	0.93
IT	I_2/I_1	0.64	4.48	2.86	0.78	5.75	4.48
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	0.71	5.21	3.72	0.78	6.05	4.72
IT	$I_d\,2f_e$	0.37	44.54	16.43	1.93	7.03	13.54
IT	$\mathrm{PI}_d\,2f_e$	0.37	44.97	16.73	1.19	11.22	13.39
IT	$V_q\,2f_e$	0.70	2.96	2.08	1.36	1.76	2.39
IT	$\lvert\Delta\bar{D}_2\rvert/D_1$	0.33	8.52	2.81	0.82	3.10	2.55
IW	V_2/V_1	0.30	2.85	0.84	0.92	0.99	0.91
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	0.31	2.40	0.74	0.88	1.10	0.97
IW	I_2/I_1	1.02	4.48	4.57	1.35	5.75	7.78
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	0.64	5.21	3.33	0.70	6.05	4.26
IW	$I_d\,2f_e$	0.51	44.54	22.66	3.88	7.03	27.29
IW	$\mathrm{PI}_d\,2f_e$	0.51	44.97	22.80	2.42	11.22	27.11
IW	$V_q\,2f_e$	0.41	2.96	1.20	0.91	1.76	1.60
IW	$\lvert\Delta\bar{D}_2\rvert/D_1$	0.27	8.52	2.34	0.84	3.10	2.59

Relative form: the SDR as defined (changes divided by the healthy level); absolute form: changes and null in the feature's own units. Num.: SCIG/PMSG ratio of the median change under fault; Den.: ratio of null quantiles $q_{95}^{\mathrm{PMSG}}/q_{95}^{\mathrm{SCIG}}$ (a quieter SCIG null raises its SDR); SDR: ratio of median SDR. IT: inter-turn; IW: inter-winding.

the PMSG the terminal-voltage group carries the strongest signature in 94 % of inter-turn and 97 % of inter-winding trials, through the voltage residual (median SDR 6.8 and 53 at five-cycle windows; 88 and 100 % of trials detectable), and the stator-current residual is weak for inter-turn faults (median SDR 1.7, 70 %). In the SCIG the controller group is strongest in 63 and 67 % of trials, through the duty-cycle residual for inter-turn faults (11.2; 93 %) and the $2f_e$ ripple of the d-axis current and its PI output for inter-winding faults (68; 100 %), the stator-current group in 34 and 33 % and the terminal-voltage group in 3 % and none. The two machines rank the 33 common features with a Spearman correlation of 0.74 (inter-turn) and 0.82 (inter-winding): the same features are informative in both, in a different order. In the WFSG the strongest inter-turn features are the terminal-voltage ratio and its residual, the current residual, the commanded q-axis voltage and the q-axis current ripple (median SDR 8–13 over all impedances), and for inter-winding faults the commanded q-axis voltage, the voltage residual and the

q-axis ripple (35); its field-current $2f_e$ ripple, an observable the other two machines lack, detected 80 % of the inter-turn trials at the matching impedance and all inter-winding trials (Table A.1). The physical quantity behind the relocation is the incremental negative-sequence impedance seen at the terminals, computed per trial from the complex residuals as $\Delta\bar{V}_2/\Delta\bar{I}_2$: 12.7 [11.1, 14.1] Ω for inter-turn and 12.2 [10.2, 13.2] Ω for inter-winding faults in the PMSG against 7.3 [6.9, 7.7] and 7.4 [7.2, 7.6] Ω in the SCIG (medians and interquartile ranges over trials, five-cycle windows), at an angle near 0° in the PMSG and near 65° in the SCIG; the magnitude-based estimate for the WFSG under inter-winding faults is 11.9 [11.0, 14.3] Ω (three-cycle windows). The PMSG drive thus converts a given negative-sequence current disturbance into 1.7 times the terminal voltage that the SCIG drive does. On the 2.5 kVA, 230 V base (21 Ω) the two impedances are 0.60 and 0.35 per unit, of the order of the leakage reactance of a passive machine or above it, so the contribution of the current regulator, which raises the closed-loop negative-sequence impedance [10, 40], cannot be separated from the machine's own without the undocumented drive parameters. Both machines run the same field-oriented controller but differ in angle source (encoder against flux observer), negative-sequence impedance and feature baselines (healthy medians of I_2/I_1, V_2/V_1 and D_2/D_1: 0.0055, 0.0009 and 0.0014 on the PMSG against 0.0079, 0.0020 and 0.0036 on the SCIG, and 0.038, 0.0067 and 0.0023 on the WFSG), so the relocation belongs to the machine–drive combination, not to the control structure alone.

5.3. The faulted phase and the extent axis

The nominal extent axis of the inter-turn test matrix carries a second variable: the four pairs of the smallest level lie in phase A, those of the middle level in phases A and B, and those of the largest level in phases A and C (Section 4.2). Table 5 and Fig. 5 show why this matters. The magnitude ratios V_2/V_1 and I_2/I_1 depend on the faulted phase far beyond what the extents explain: within the 11.6–12.1 % level, the two phase-A pairs give V_2/V_1 median SDRs of 3.4 and 13.8 on the PMSG against 17.2 and 26.3 for the two phase-C pairs, and the spread between the pairs of one level is a factor 2.7 to 14 for V_2/V_1 and 1.3 to 7 for I_2/I_1. The phasor residuals remove this almost entirely: within every level the pairs agree to within a factor 1.1 to 1.8 on both machines (17.3–22.8 for $|\Delta\bar{V}_2|/V_1$ in the 11.6–12.1 % level of the PMSG), and the per-phase medians of the residuals follow the extents of the pairs each phase holds. The mechanism is the one of Section 4.3: the inherent

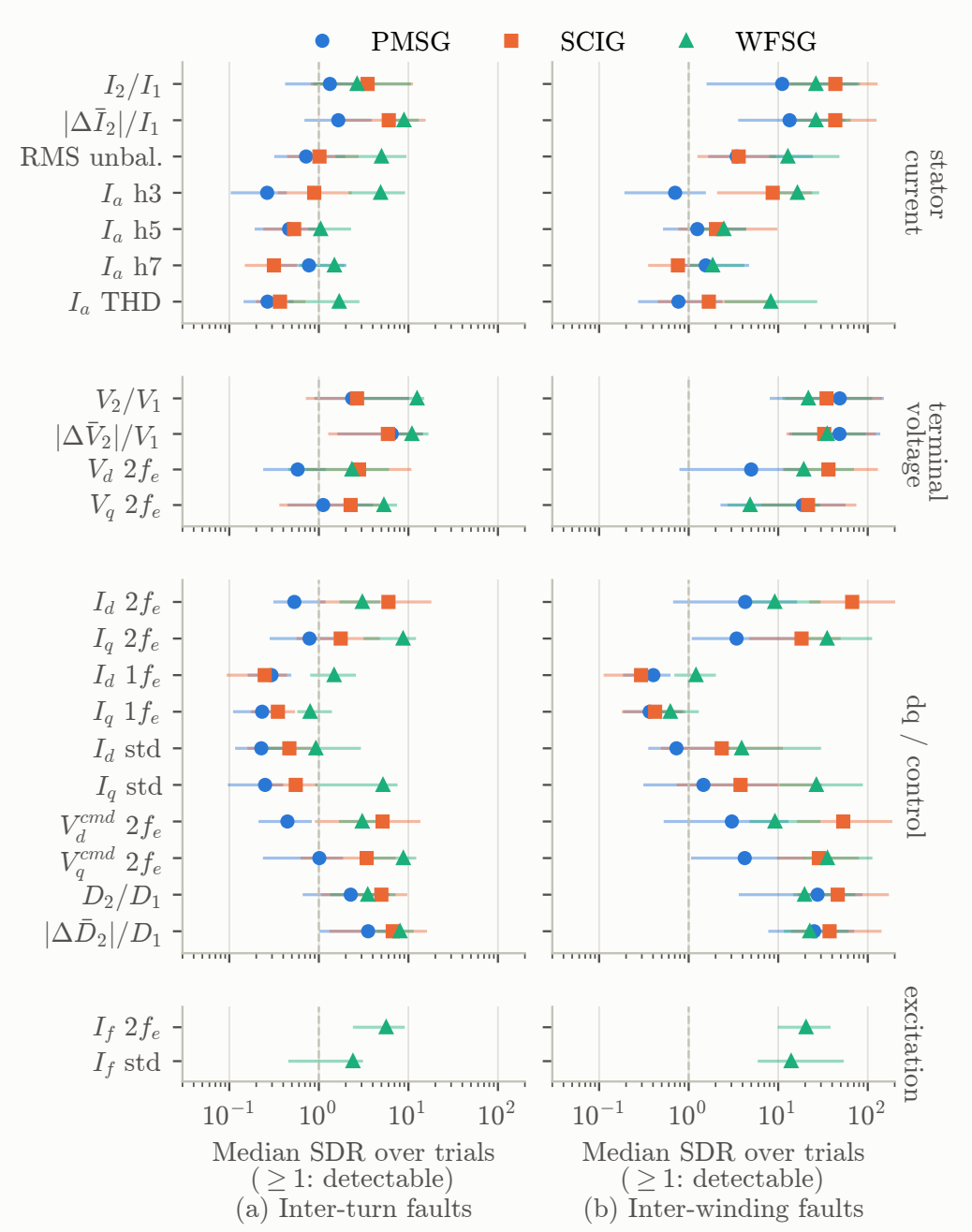


Figure 4: Median SDR of every feature (dot) with interquartile range (bar) for the three alternatives, three-cycle windows; the WFSG is pooled over its fault impedances

Table 5: Inter-turn median SDR / detectable share (%) by faulted phase, five-cycle windows

Machine	Phase (pairs)	Extents (%)	V_2/V_1	$\lvert\Delta\bar{V}_2\rvert/V_1$	I_2/I_1	$\lvert\Delta\bar{I}_2\rvert/I_1$	$I_d\,2f_e$
PMSG	A (8)	2.7, 2.8, 3.8, 7.4, 11.6, 12.0	1.3 / 57	3.3 / 82	0.76 / 40	1.3 / 56	0.31 / 10
PMSG	B (2)	7.4, 7.7	2.3 / 83	6.8 / 100	1.6 / 78	1.8 / 100	0.32 / 22
PMSG	C (2)	11.6, 12.1	20 / 100	19 / 100	5.3 / 89	5.3 / 100	1.5 / 83
SCIG	A (8)	2.7, 2.8, 3.8, 7.4, 11.6, 12.0	1.1 / 54	3.2 / 72	1.2 / 57	2.9 / 78	1.5 / 64
SCIG	B (2)	7.4, 7.7	6.4 / 100	7.2 / 100	5.3 / 100	6.3 / 100	11 / 100
SCIG	C (2)	11.6, 12.1	23 / 100	22 / 100	18 / 100	19 / 100	36 / 100

Number of tap pairs in parentheses; extents: design-level extents of those pairs (%). The magnitude ratios V_2/V_1 and I_2/I_1 depend on the angle between the inherent and the fault-induced asymmetry, which rotates with the faulted phase; the phasor residuals $\lvert\Delta\bar{V}_2\rvert/V_1$ and $\lvert\Delta\bar{I}_2\rvert/I_1$ remove that angle.

negative-sequence component of each machine (healthy V_2/V_1 of 0.0009 on the PMSG and 0.0020 on the SCIG) is of the same order as the fault-induced one for a small inter-turn fault, so the magnitude of their sum rises or falls with the angle between them, which rotates by $\pm 120°$ with the faulted phase; the complex change does not. Two consequences follow for the rest of the paper. First, the residual features replace the magnitude ratios wherever a sequence quantity represents a suite (Tables 6–8, Fig. 6); the magnitude ratios remain in Fig. 4 and Table 4 because they are the established indicators. Second, on the residual features the two machines on the common bench are nearly indistinguishable through the terminal voltage—per-level medians of $\lvert\Delta\bar{V}_2\rvert/V_1$ of 1.1, 6.8 and 19.5 (PMSG) against 0.9, 6.9 and 19.5 (SCIG) at the three inter-turn levels—and differ through the stator current, where the SCIG's residual is 2 to 4 times the PMSG's (0.5, 1.7 and 5.1 against 1.1, 6.3 and 17.3): the relocation of the signature survives the change of feature, the earlier reading that the PMSG's voltage signature is stronger than the SCIG's does not.

5.4. Minimum detectable extent

Table 6 gives, for the two residual sequence features, the d-axis PI ripple and the q-axis voltage ripple, the MDE as a bracket on the physical levels, the detectable share and the median SDR with intervals from the tap-pair bootstrap in which the null is re-estimated, and the SCIG/PMSG ratio of medians; Fig. 6 plots the SDR of the two residual features against the nominal extent (left; dots are trials, large markers the median per tap pair, their shape

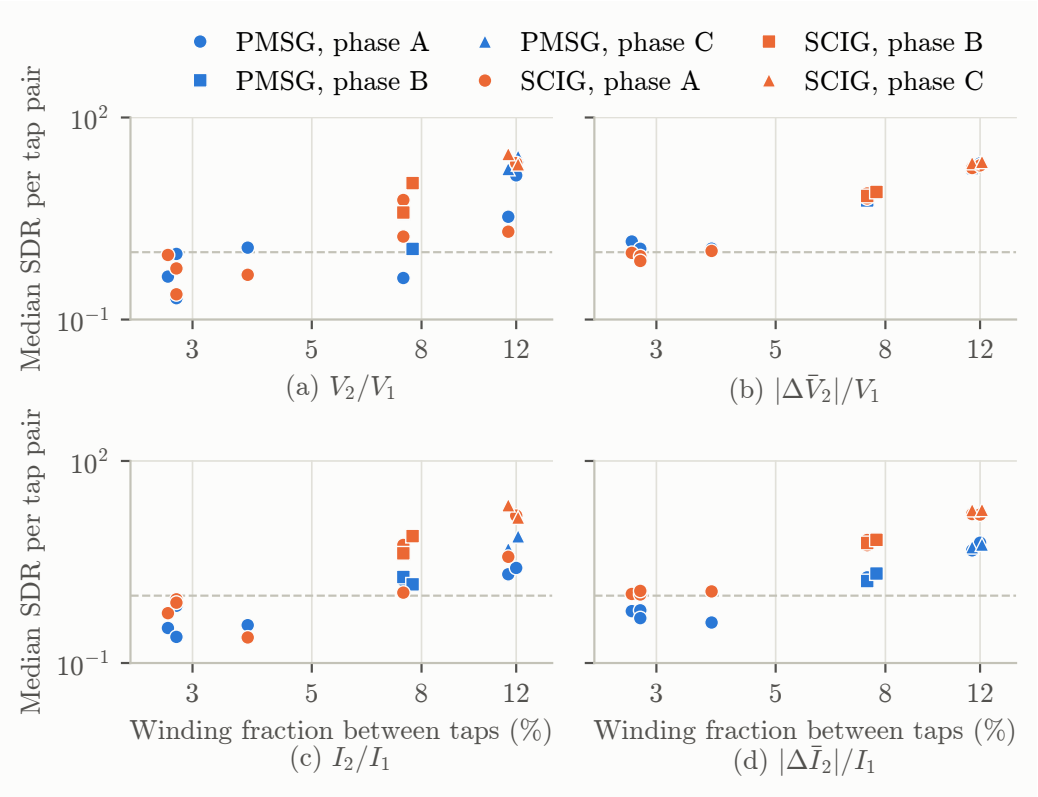


Figure 5: Inter-turn median SDR per tap pair against the nominal extent: magnitude ratios (**a**, **c**) and phasor residuals (**b**, **d**); marker, faulted phase; colour, machine

the fault impedance) and against the physical severity axis (right; RMS fault current relative to the RMS pre-fault stator current, all trials). Through the terminal-voltage residual the PMSG detects every inter-turn level including the smallest (2.7–3.8 %; per-level median SDR 1.1, share 88 % [74, 98]) and the SCIG every level from the middle one (bracket (2.7–3.8, 7.4–7.7]; its smallest level sits at a median of 0.9); through the stator-current residual the roles reverse, the SCIG detecting every level and the PMSG from the middle one; the d-axis ripple detects from the middle level on the SCIG and only the largest level on the PMSG. Where a machine detects the smallest level its per-level median is within 15 % of one, so these brackets sit at the limit of the test matrix, which the bootstrap intervals say (the smallest or the middle level in every such case). For inter-winding faults the PMSG detects from the smallest level (2.3–3.2 %) through the voltage residual, with every trial detected, but only from 9.7–10.6 % through the current residual and from 14.3–14.8 % through the d-axis ripple, while the SCIG detects from the smallest level on every feature except the q-axis voltage ripple. The SDR rises with extent in every machine (Spearman 0.51–0.95). The SCIG/PMSG ratio of median SDR, the quantity best supported by twelve tap pairs, is 1.0 [0.7, 1.3] for the voltage residual and 3.7 [2.1, 5.1] for the current residual on inter-turn faults, 0.7 [0.6, 1.1] and 3.3 [2.1, 6.6] on inter-winding faults, and 16 [3, 42] for the d-axis ripple: the two machines are indistinguishable through the terminal voltage, differ by a factor three to four through the stator current and by an order of magnitude through the controller. On the physical axis (Fig. 6, right) the picture is the same and independent of the tap-pair design: at equal fault-to-stator current ratio the SCIG's current residual lies above the WFSG's, which lies above the PMSG's, while on the voltage residual the three machines nearly coincide. For the WFSG, whose small inter-turn extents were tested only at the lower impedance, the MDE at the matching impedance is bounded by the smallest level tested there (11.6–12.1 %, detectable in 80–92 % of trials depending on the suite); its 2.7–7.7 % inter-turn trials at 1 Ω sit at SDR ≈ 1 on the sequence features, and all its inter-winding extents from 2.3 % are detectable.

5.5. Operating envelope

Speed dominates the operating-point dependence. For the fixed features of the suite tables the median SDR of the inter-turn trials rises between 1200 and 1800 rpm from 5.3 to 10.0 (voltage residual) and from 2.8 to 5.6 (duty-cycle residual) on the PMSG, and from 3.8 to 16.6 (current residual) and

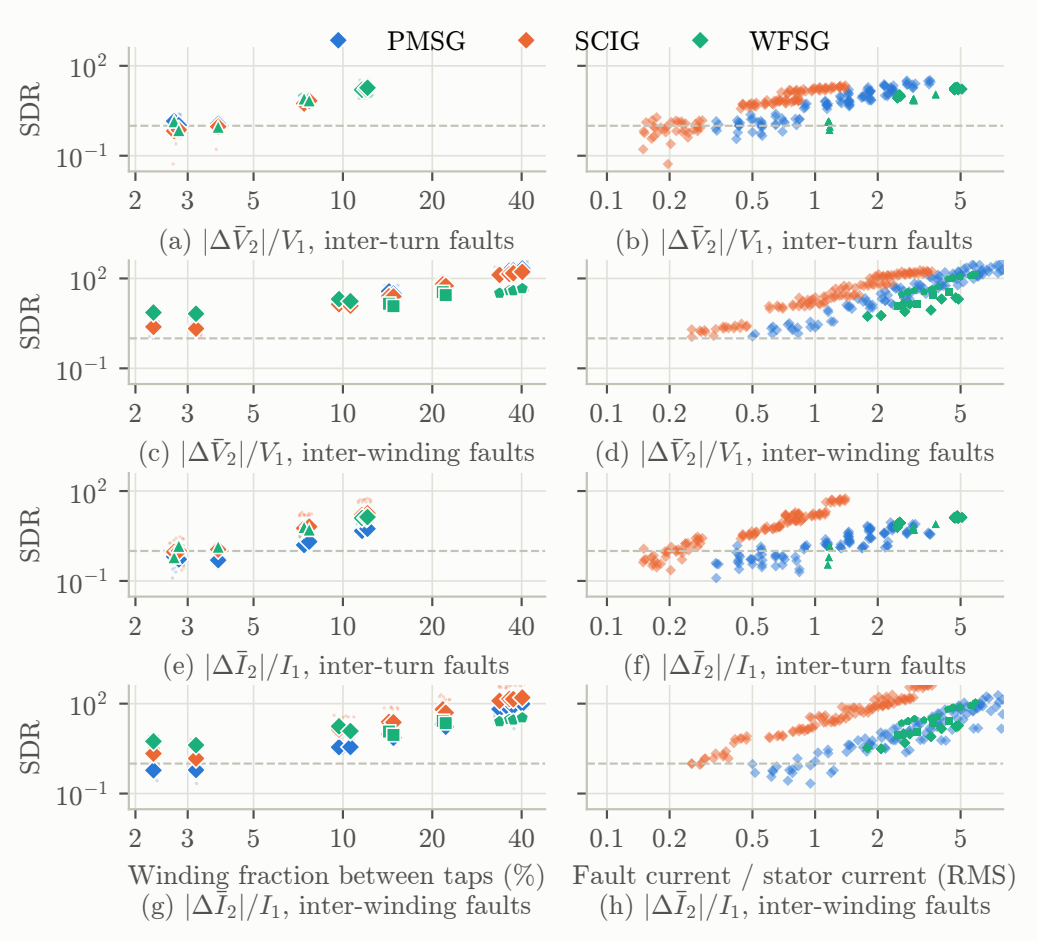


Figure 6: SDR of the voltage residual (**a**–**d**) and the current residual (**e**–**h**) against the nominal extent (left) and the physical severity (right); dashed line, SDR = 1

Table 6: Minimum detectable extent, detectable share and median SDR with 95 % intervals, four features, PMSG and SCIG, five-cycle windows

Class	Feature	M.	MDE	DS [CI]	Median SDR [CI]	S/P ratio [CI]
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	P	≤3.0	88 [74, 98]	6.84 [1.67, 15.9]	
		S	(3.0, 7.5]	81 [63, 95]	6.9 [1.16, 16.4]	1.01 [0.683, 1.29]
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	P	(3.0, 7.5]	70 [44, 93]	1.7 [0.751, 4.14]	
		S	≤3.0	85 [70, 96]	6.34 [1.93, 13.2]	3.72 [2.08, 5.08]
IT	$\mathrm{PI}_d\,2f_e$	P	(7.5, 11.8]	24 [7, 44]	0.37 [0.289, 0.797]	
		S	(3.0, 7.5]	76 [57, 93]	6.18 [1.22, 18.8]	16.7 [3.04, 41.7]
IT	$V_q\,2f_e$	P	(7.5, 11.8]	51 [31, 73]	1.13 [0.649, 3.96]	
		S	(3.0, 7.5]	60 [34, 84]	2.34 [0.47, 5.07]	2.08 [0.492, 3.35]
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	P	≤2.8	100 [100, 100]	52.7 [15.3, 148]	
		S	≤2.8	100 [99, 100]	39 [14.9, 136]	0.74 [0.607, 1.1]
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	P	(2.8, 10.1]	86 [68, 100]	13.7 [4.1, 48.3]	
		S	≤2.8	99 [95, 100]	45.8 [16.5, 104]	3.33 [2.08, 6.64]
IW	$\mathrm{PI}_d\,2f_e$	P	(10.1, 14.6]	69 [49, 88]	3 [0.937, 10.1]	
		S	≤2.8	100 [100, 100]	68.4 [23.6, 223]	22.8 [14.2, 36.7]
IW	$V_q\,2f_e$	P	(2.8, 10.1]	89 [77, 98]	18.7 [4.23, 55.1]	
		S	(2.8, 10.1]	88 [71, 100]	22.5 [7.26, 68.4]	1.2 [0.76, 1.97]

$\alpha = 0.95$. MDE on the physical extent levels (% of a branch): $(a, b]$ means the level a is not detected and every level from b up is; $\leq b$: every tested level is detected. DS (%), median SDR and the SCIG/PMSG ratio of medians carry percentile intervals from a bootstrap over tap pairs (the same pairs for both machines) in which the null quantile is re-estimated in every resample. IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG.

from 9.5 to 12.2 (duty-cycle residual) on the SCIG, with detectable shares rising from 78 to 97 and from 69 to 83 % on the PMSG and from 69 to 100 and from 86 to 97 % on the SCIG; the PMSG's current residual is the exception, flat at 1.4–2.6 with shares of 67–78 %. Torque matters less: at a fixed speed the median of the best key feature changes by less than a factor 1.25 between 5.2 and 8.0 N m. Two cautions apply. The dependence is consistent with a resistor-limited fault current that scales with speed, so it is a property of the emulation as much as of the machines. And the pooled null is a mixture over operating points, so the 5 % false-alarm reading holds on average over the envelope: the share of trials whose healthy change exceeds the pooled 95th percentile is 0 to 16 % by speed for the key features, the extremes being the SCIG's current residual at 1800 rpm (16 %; its electrical frequency of 59.6 Hz then nearly coincides with the 60 Hz grid, a possible source of a slow beat within the windows) and the PMSG's current residual at 1500 rpm (15 %). Both make the lowest monitored speed a parameter of the requirement, and a field monitor would keep a null per speed band.

5.6. Sensor suites and the selection

Table 7 gives the suite results for the PMSG and SCIG at five-cycle windows, Table A.1 (Appendix A) for the three alternatives on the common feature set at three cycles, and Table 8 the decision table of step 5. For the PMSG, current transducers alone detect inter-turn faults from the middle level (bracket (3.0, 7.5]) and inter-winding faults from 9.7–10.6 %; three voltage transducers take both to the smallest level through the voltage residual (88 and 100 % of trials), and neither controller access nor a torque transducer improves the fixed-feature result further. The drive-internal suite reaches the same brackets as the current transducers through the duty-cycle residual, with a higher share for inter-winding faults (96 %). For the SCIG the cheapest suites already reach what any suite reaches: at five cycles the smallest level of both classes, through the current residual or, with no added sensor, the duty-cycle residual (93 % of inter-turn trials) and the d-axis PI output (100 % of inter-winding trials), so additional hardware changes the fixed feature but not the reach; at three cycles its two cheapest inter-turn suites stop at the middle level (Table A.1). For the WFSG every suite detects all inter-winding faults from 2.3 % and 92 % of the inter-turn trials at the matching impedance, the field current alone 100 and 80 % (its inter-turn MDE there is bounded by the smallest tested extent, 11.6 %). The fixed features have at most two false alarms in nine healthy trials (the SCIG's duty-cycle residual), and the nested selection changes no share by more than six points. The calibrated oracle, the per-trial best feature at the threshold that holds the false-alarm rate of the maximum at 5 %, exceeds the fixed feature's share only for the PMSG's current-only suite on inter-winding faults (89 against 86 %) and otherwise equals it or falls below it by up to 22 points, so combining the features of a suite buys nothing over its best feature once the false-alarm rate is held; the uncalibrated per-trial maximum reaches higher shares at false-alarm rates of 56–78 % on the healthy trials for the drive-internal and the largest suites. A shaft torque transducer adds nothing for winding faults on the two machines on which it was available: the mechanical-torque features' healthy relative change is unbounded on the SCIG (their healthy level crosses zero) and their median SDR is below 0.3 on the PMSG.

The decision table (Table 8) turns this into the selection of step 5, with the requirement evaluated at the tested levels and a required detectable share of 0.9 above it. For inter-turn faults at the smallest level (2.7–3.8 % of a branch) no suite on either machine meets the requirement at $\alpha = 0.95$: the PMSG's voltage suite (probability 0.27) and the SCIG's drive-internal suite (0.82) are

Table 7: Sensor suites versus what they detect, PMSG and SCIG, five-cycle windows: fixed feature, MDE bracket, detectable share and false alarms

Suite	PMSG				SCIG			
	feature	MDE	DS (oracle)	FA	feature	MDE	DS (oracle)	FA
Inter-turn faults								
drive	$\lvert\Delta\bar{D}_2\rvert/D_1$	(3.0, 7.5]	75 (69)	0/9	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤3.0	93 (71)	2/9
3CT	$\lvert\Delta\bar{I}_2\rvert/I_1$	(3.0, 7.5]	70 (61)	0/9	$\lvert\Delta\bar{I}_2\rvert/I_1$	≤3.0	85 (78)	1/9
3CT+3VT	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤3.0	88 (75)	0/9	$\lvert\Delta\bar{I}_2\rvert/I_1$	≤3.0	85 (77)	1/9
3CT+3VT+drv	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤3.0	88 (69)	0/9	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤3.0	93 (74)	2/9
+torque	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤3.0	88 (69)	0/9	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤3.0	93 (74)	2/9
Inter-winding faults								
drive	$\lvert\Delta\bar{D}_2\rvert/D_1$	≤2.8	96 (90)	0/9	$\mathrm{PI}_d\,2f_e$	≤2.8	100 (100)	1/9
3CT	$\lvert\Delta\bar{I}_2\rvert/I_1$	(2.8, 10.1]	86 (89)	0/9	I_2/I_1	≤2.8	99 (98)	0/9
3CT+3VT	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤2.8	100 (98)	0/9	V_2/V_1	≤2.8	100 (99)	0/9
3CT+3VT+drv	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤2.8	100 (98)	0/9	$\mathrm{PI}_d\,2f_e$	≤2.8	100 (100)	1/9
+torque	$\lvert\Delta\bar{V}_2\rvert/V_1$	≤2.8	100 (96)	0/9	$\mathrm{PI}_d\,2f_e$	≤2.8	100 (100)	1/9

Suites are cumulative (drive: drive-internal quantities; 3CT: three current transducers; +3VT: three voltage transducers added; +drv: controller access added; +torque: shaft torque transducer added). DS (%) of the fixed feature, in parentheses the share of the calibrated oracle (per-trial best feature at the threshold that holds the false-alarm rate of the maximum at 5 %); FA: false alarms of the fixed feature on the nine healthy trials. MDE as in Table 6; features failing the reliability screen are excluded for that machine.

uncertain, because their shares at that level fall short of 0.9 in most resamples, and the current-only suites fail. From the middle level (7.4–7.7 %) every suite on both machines meets it with probability 1.00, so no added sensor is needed for inter-turn faults of that size on either machine; on the WFSG only the largest level is testable at the matching impedance, and there every suite meets it except the field current alone (0.92; its share is 80 %). For inter-winding faults the SCIG and the WFSG meet the requirement at the smallest level with every suite including the drive-internal one, whereas the PMSG needs voltage transducers there (drive-internal 0.93, current-only fails) and from 9.7–10.6 % meets it with every suite (current-only 0.97). Relaxing α to 0.90 admits the SCIG's drive-internal suite at the smallest inter-turn level (0.98) and the PMSG's drive-internal suite at the smallest inter-winding level (1.00) but makes the PMSG's current-only suite uncertain for inter-winding faults at every level (0.74–0.87); tightening it to 0.99 turns the uncertain inter-turn verdicts at the smallest level into failures on both machines and makes the PMSG's current-only suite uncertain at the middle inter-turn level (0.53). On three-cycle windows the SCIG's inter-turn verdicts at the smallest level are failures for every suite, the window dependence noted in Section 5.8.

Table 8: Decision table at $\alpha = 0.95$: bootstrap probability that a suite meets the requirement at each tested extent level

Suite	PMSG				SCIG				WFSG			
	probability of meeting the requirement at e_{req} (%)											
Inter-turn faults												
e_{req} (%)	3.0	7.5	11.8		3.0	7.5	11.8		3.0	7.5	11.8	
drive	0.00	**1.00**	**1.00**		0.82	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
3CT	0.00	**1.00**	**1.00**		0.05	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
3CT+3VT	0.27	**1.00**	**1.00**		0.05[b]	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
3CT+3VT+drv	0.27[b]	**1.00**	**1.00**		0.83	**1.00**	**1.00**		n.t.	n.t.	**1.00**	
exc	–	–	–		–	–	–		n.t.	n.t.	0.92	
Inter-winding faults												
e_{req} (%)	2.8	10.1	14.6	21.9	2.8	10.1	14.6	21.9	2.8	10.1	14.6	21.9
drive	0.93	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
3CT	0.00	**0.97**	**0.97**	**0.99**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
3CT+3VT	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
3CT+3VT+drv	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	**1.00**	n.t.	n.t.
exc	–	–	–	–	–	–	–	–	**1.00**	**1.00**	n.t.	n.t.

Meets: MDE $\leq e_{\text{req}}$ and detectable share ≥ 0.9 for $e \geq e_{\text{req}}$, with e_{req} at the physical levels of the common bench (% of a branch; the WFSG is assessed against the same values on its own tested extents); 1000 resamples of the operating-point trials within each level, the null quantile and the fixed feature re-estimated in every resample. Bold: probability ≥ 0.95 (meets); below 0.05 the suite fails; n.t.: not testable, the machine has no tested extent on the required side of e_{req}. PMSG and SCIG at five-cycle windows, WFSG at three cycles and at its matching fault impedance. The inter-winding levels 35.9 and 40.2 % are omitted: every suite meets them. Suites as in Table 7; exc: WFSG field current.

[b] Probability inherited from a cheaper subset of the suite (a suite may always use the features of its subsets).

Read as a specification, the table says that an inter-turn requirement at the smallest tested level cannot be met with confidence by any suite on these benches, that a requirement at the middle level is met by the cheapest suite on all three machines, and that the inter-winding requirement at the smallest level is the one case in which the sensor decision depends on the machine type.

5.7. Transfer across alternatives

Table 9 reports step 6 for the PMSG–SCIG pair on five-cycle windows and Table A.5 (Appendix A) for the three alternatives on three-cycle windows. All figures are on the out-of-reference windows (those on all windows agree with them to within 0.01), the intervals are 95 % cluster-bootstrap intervals over the tested machine's trials, and the true-positive rate (TPR) is read

at 1 % false-positive rate. In physical units the GBDT detector reaches an area under the receiver operating characteristic curve (AUC) of 0.93 [0.86, 0.98] and 0.91 [0.83, 0.97] within a machine on unseen tap pairs but 0.79 [0.68, 0.88] (PMSG→SCIG) and 0.61 [0.50, 0.72] (SCIG→PMSG) across, with a TPR of 0.25 and 0.26 across against 0.74 and 0.67 within. Referencing the features to healthy behaviour of the target machine recovers most of the AUC under all three healthy-referenced rules: 0.88 [0.81, 0.95] and 0.88 [0.81, 0.94] with the features scaled on all healthy trials of the target, 0.87 [0.79, 0.96] and 0.88 [0.80, 0.94] scaled on the healthy trial at the same operating point, and 0.93 [0.87, 0.98] and 0.89 [0.82, 0.95] referenced to the first half of each trial's own reference interval, against 0.94 and 0.96 within a machine under that rule. The rules separate on the TPR and on the linear detector. Across the machines the self-referenced rule gives a TPR of 0.81 [0.67, 0.94] (PMSG→SCIG) and 0.54 [0.37, 0.69] (SCIG→PMSG), the per-operating-point baseline 0.71 [0.57, 0.87] and 0.62 [0.42, 0.76] and the all-healthy baseline 0.50 [0.36, 0.67] and 0.54 [0.32, 0.71], so the self-referenced rule is the best on AUC in both directions and on TPR in one, with overlapping intervals in the other; and a logistic regression transfers at 0.93 and 0.91 when self-referenced but at 0.78 and 0.64 under the all-healthy baseline and at 0.81 and 0.50 under the per-operating-point baseline. The threshold that gives 1 % false positives on all negatives fires on 0 to 5 % of the healthy trials' fault-time windows, the only negatives that contain the contactor operation. Between the three alternatives, self-referenced GBDT detectors transfer in all six directions with an AUC of 0.85 [0.79, 0.91] (SCIG→PMSG) to 0.98 [0.96, 1.00] (PMSG→WFSG) and a TPR of 0.41 [0.25, 0.54] to 0.94 [0.86, 0.99], against 0.90–0.97 and 0.71–0.93 within a machine, and a detector trained on two machines detects the faults of the third with an AUC of 0.87 [0.81, 0.93] (PMSG), 0.91 [0.85, 0.96] (SCIG) and 0.99 [0.97, 1.00] (WFSG; optimistic, as that bench has no healthy trials). Restricted to each suite's features, every suite transfers between the PMSG and the SCIG at 0.87–0.95 when self-referenced, at 0.74–0.90 under the all-healthy baseline and at 0.83–0.89 under the per-operating-point baseline.

5.8. Robustness and the two nulls

Table A.2 (Appendix A) shows how the tuning choices move the design quantities, with the screen applied at every setting. Across window lengths of three, five and eight cycles the MDE brackets of the key features are unchanged on both machines, with three exceptions at the margin: the SCIG's

Table 9: Transfer of a GBDT detector between PMSG and SCIG under four referencing rules, out-of-reference windows, 95 % intervals

Direction	Metric	physical units	z, all healthy trials of the target	z, healthy trial at the operating point	$\lvert z\rvert$ against own reference
P→P	AUC	0.93 [0.86, 0.98]	0.93 [0.86, 0.98]	0.91 [0.84, 0.96]	0.94 [0.90, 0.98]
	TPR	0.74 [0.55, 0.89]	0.74 [0.55, 0.89]	0.63 [0.43, 0.83]	0.78 [0.64, 0.91]
S→S	AUC	0.91 [0.83, 0.97]	0.91 [0.83, 0.97]	0.91 [0.84, 0.96]	0.96 [0.92, 0.99]
	TPR	0.67 [0.50, 0.85]	0.67 [0.50, 0.85]	0.62 [0.43, 0.79]	0.82 [0.68, 0.94]
P→S	AUC	0.79 [0.68, 0.88]	0.88 [0.81, 0.95]	0.87 [0.79, 0.95]	0.93 [0.87, 0.98]
	TPR	0.25 [0.14, 0.35]	0.50 [0.36, 0.67]	0.71 [0.57, 0.87]	0.81 [0.67, 0.94]
S→P	AUC	0.61 [0.50, 0.72]	0.88 [0.81, 0.94]	0.88 [0.80, 0.94]	0.89 [0.82, 0.95]
	TPR	0.26 [0.10, 0.40]	0.54 [0.32, 0.71]	0.62 [0.42, 0.76]	0.54 [0.37, 0.69]

AUC and true-positive rate at 1 % false-positive rate (TPR) on the out-of-reference windows (those outside the normalising set, Section 4.4); intervals from a bootstrap over the trials of the tested machine (500 resamples of its tap pairs and healthy groups). P→P, S→S: cross-validation grouped by tap pair; P→S, S→P: trained on one machine, tested on the other.

current residual reaches the smallest inter-turn level at five and eight cycles but only the middle one at three, the PMSG's q-axis voltage ripple reaches the middle level only at eight, and the PMSG's d-axis features, whose null quantile exceeds one at eight cycles, are screened out there. The detectable shares move by at most 7 points, except those of the PMSG's d-axis features, which lie below SDR = 1 at every setting (13 points). Tightening the null quantile from 0.95 to 0.99 lowers the shares by 1 to 14 points, moves the MDE up one level for the PMSG's voltage residual and the SCIG's current residual and screens out the PMSG's d-axis features (null quantile above one); relaxing it to 0.90 raises the shares by up to 10 points and moves the q-axis voltage ripple down one level on both machines and the SCIG's voltage residual and d-axis ripple down one level. The orderings of the two machines on the residual and d-axis features are the same at every setting; on the terminal-voltage magnitude ratio the two machines are tied within three points and swap order with α. The fixed-feature brackets of Table 7 are the same at three and five cycles on the PMSG, whereas the SCIG's two cheapest inter-turn suites reach the smallest level at five cycles and only the middle one at three (Table A.1), which is why the SCIG's inter-turn verdict at the smallest level depends on the window length (Section 5.6). At $\alpha = 0.95$ the binomial rule of the screen removes one feature, the duty-cycle magnitude ratio on the SCIG (three false alarms in nine healthy trials at three and five cycles), and the null-quantile rule removes the torque features and, at eight

cycles, the PMSG's d-axis features (Appendix A.2); with nine trials the binomial rule has a size of 0.8 % per feature, so 0.3 chance exclusions among the 33 features of a machine are expected, and the largest false-alarm count among the key features is two of nine (the SCIG's terminal-voltage ratio at three cycles and its d-axis features at $\alpha = 0.90$). An absolute-change variant of the SDR, which keeps the healthy level of the feature out of numerator and null, ranks the 33 features with a Spearman correlation of 0.92–0.99 against the relative variant in the four machine–class combinations and shares two of its three top features in each. The tap-pair bootstrap puts the median SDRs within a factor of two to four and the inter-turn shares within about ±15 points (Table 6); the null quantile itself has a sampling interval spanning a factor 1.3 to 1.5 (Table A.4), which these intervals include.

Table A.3 (Appendix A) replaces the reference immediately before the fault by the healthy trial recorded at the same operating point at another time, a commissioning baseline: the null of the four key features widens by a factor of 1.1 to 4.6 (1.9 to 3.2 for the voltage and current residuals), the median SDR of the inter-turn residuals falls from 6.8 to 2.4 (PMSG, voltage) and from 6.3 to 2.1 (SCIG, current) and their detectable shares from 88 and 85 % to 72 and 69 %, while inter-winding faults remain detectable in 67–93 % of trials. The MDEs reported here are therefore statements about a short-horizon change detector; a monitor that compares against an old baseline needs the wider null, and Section 5.7 shows what the choice of reference costs a learned detector.

6. Discussion

6.1. What the demonstration establishes about the method

The demonstration supports four claims about the method that do not depend on which machine came out ahead, and it corrects the naive form of three of them.

The design-level description of a failure mode does not fix its physical severity, and the test matrix can carry a hidden variable. A short-circuit of the same winding fraction produced a fault current that differed by about 10 % between the PMSG and the SCIG in amperes and by a factor of 2.3 relative to the stator current the sensors see; on the WFSG bench extent and fault impedance were confounded; and on all three benches the faulted phase rode on the nominal extent axis, which the phasor residuals remove and the magnitude ratios do not (Section 5.3). A method that compares

alternatives on a failure-mode class must therefore carry two severity axes—the design-level one, on which requirements are written, and a physical one, on which the alternatives are compared—state which one a number refers to, and use features that are invariant to the nuisance variables of the matrix; for resistor-limited emulation the physical axis is predictable at the design stage from volts per turn, speed and the resistance.

Where a failure becomes observable is a property of the machine–drive combination, and the SDR must be read with its decomposition in both forms. In the PMSG the winding asymmetry appeared in the terminal voltage, in the SCIG in the controller quantities and the stator current, in the WFSG additionally in the field current; the incremental negative-sequence impedance under fault was 1.7 times larger in the PMSG drive than in the SCIG drive. The version of this claim in which "the control structure decides" is not supported: both machines run the same controller, and on the residual features they are indistinguishable through the terminal voltage. Nor is the version in which the SCIG wins only by a quieter null: Table 4 shows that this holds for the stator current, where the PMSG's signature is the larger one in amperes, but that for the controller quantities the SCIG has the larger signature as well once the features are read in absolute units. The SDR is designed to credit a quiet alternative, but a design team must know whether an alternative wins by signature or by silence, because the two generalise differently: a signature is a property of the fault physics, a null of the sensing and control chain and of the operating regime, which a design-stage instantiation cannot simulate (Section 3.5), and the decomposition in the relative and the absolute form is the way to tell.

The cheapest adequate sensor suite depends on the alternative and on the requirement, and here it would change the monitoring specification, not the choice of machine type. With three current transducers the PMSG detects inter-turn faults from the middle tested level (7.4–7.7 % of a branch) and inter-winding faults from 9.7–10.6 %; three voltage transducers take both to the smallest level and buy nothing on the SCIG, whose current-only or drive-internal suites already reach it. The decision table makes the requirement explicit: at the middle inter-turn level all three alternatives meet it with drive-internal quantities and diagnosability does not discriminate between them; at the smallest level no suite on any bench meets it with the stated power, so the question is answered "not with these data" rather than "with voltage transducers"; and for inter-winding faults at the smallest level the PMSG is the one alternative that needs added sensors. The method would

change the choice of machine type only if, for one alternative, no affordable suite met the requirement; here it changes what is specified for each: a measured detectability per alternative and suite, with its uncertainty, in place of an assumed one.

A monitoring design transfers across alternatives under a healthy-referenced rule, and the rule belongs in the specification. A detector built on one machine and applied to another in physical units lost much of its performance (AUC 0.61–0.79 across against 0.91–0.93 within); scaling its features on healthy trials of the target machine, all of them or the one at the operating point, recovered most of it (0.87–0.88); and referencing each window to its own trial's preceding healthy interval gave the highest AUC in both directions (0.89–0.93) and transferred the three machines in all six directions and from any two to the third. The healthy-referenced rules differ in the true-positive rate at low false-alarm rates and in the linear detector, which transfers in both directions only when self-referenced; the earlier form of this claim, that a detector transfers only when self-referenced, is not supported by the intervals. The between-trial null (Table A.3) explains the ordering: against a baseline recorded at another time the inter-turn signatures sit at an SDR of about 2 instead of 6 to 7, so the self-referenced rule sees the cleaner signal, at the price of being an abrupt-onset detector that a slowly growing fault would evade. For a product family the specification should therefore name the referencing rule and the horizon over which the reference is renewed, and a monitor that compares against an old baseline must be calibrated on the wider null.

6.2. Implications for the condition monitoring of wind-turbine generators

Within the scope of Section 6.3, the results carry the following implications for the wind-turbine generator. A PMSG with full-power conversion should be specified with terminal voltage measurement, or with another observable of the terminal asymmetry such as the zero-sequence voltage where a neutral is accessible [42], if inter-winding faults of the smallest tested size (2.3–3.2 % of a branch) are to be detected by threshold monitoring with the stated power; for inter-turn faults the voltage measurement raises the detectable share at the smallest level from 70–75 to 88 % without making that level a confident requirement, and from the middle level the controller's duty-cycle residual suffices. An SCIG reaches the same detectability with current transducers alone or with no added sensor, and additional hardware buys robustness rather than reach. A wound-field machine with a static

exciter offers a further drive-internal channel, the field current, whose $2f_e$ ripple detected every inter-winding fault and 80 % of the inter-turn trials at the matching impedance; a brushless exciter would not expose it [41]. A shaft torque transducer, as filtered on these benches, is not worth specifying for winding faults. Two industrial parameters change these statements: the ratio of current-loop bandwidth to twice the electrical frequency, which for a direct-drive multi-megawatt PMSG at a fundamental near 10 Hz is expected to push the signature further into the controller and the voltage than in these 40–60 Hz machines, although the current-loop gains and bandwidths of the benches are not documented in the data papers and the statement is a mechanism, not a computed prediction; and the grid-connected, unregulated stator of a doubly-fed generator, in which the stator and rotor currents carry the signature that the rotor-current and search-coil methods of Shah et al. [57] exploit, so that whether its machine–drive combination relocates the signature at all is not established by these data.

6.3. Limitations

The most important limitation is that each alternative is represented by one physical machine. The differences reported are differences between three machines; attributing them to topology rather than to the individual units requires replicated machines or simulated populations of each alternative under parameter variation, validated against the measured units. The demonstration is therefore regarded as establishing the method and the mechanisms, not the population-level ranking of the topologies; the two benches also differ in converter, controller, sampling rate and sensing chain.

The fault emulation is resistor-limited and the benches are laboratory scale (2–2.5 kVA): a bolted field fault is limited by the leakage reactance of the shorted turns and reaches several times the rated current, so its signature-versus-extent law, and hence every MDE, differs in kind from the one measured here, and the smallest tested fault (2.7–3.8 % of a branch) is larger than the one- to three-turn fault that starts an insulation failure in a large winding. The operating points were scaled from a 15 MW turbine by the speed ratio only (1800 rpm at the generator for 7.55 rpm at the rotor, the torque defined analogously); the electrical frequency, the per-unit reactances and the ratio of current-loop bandwidth to twice the electrical frequency are those of the benches, whose controller gains and bandwidths the data papers do not state, so the relocation of the signature cannot be computed for another drive and Section 6.2 is conditional. Only winding faults and only electrical

and controller observables were studied at a fixed thermal state and without nuisance transients other than the contactor operation; the suites are those the datasets can populate (Section 4.3), and the drive-internal suite that the decision table often finds cheapest is available to an OEM-integrated monitor but not to a third-party retrofit, for which three current transducers are the cheapest option. On the WFSG the fault impedance is confounded with extent; only four tap pairs per class exist at the impedance matching the other two machines, so its inter-turn MDE is bounded, not measured; it has no healthy trials, so the reliability screen cannot be applied to it and its transfer entries lack nuisance-transient negatives; and 81 of its 438 trials hold fewer than three three-cycle windows. The PMSG/SCIG faulted interval contains about 55 ms of healthy time before the relay picks up, which dilutes those SDRs slightly relative to the WFSG's.

Statistically, the pooled null is a mixture over operating points and holds its nominal false-alarm rate on average, not per speed; the screen rests on nine healthy trials and has little power; the fixed feature is chosen on the trials it is evaluated on, which the nested variant shows to move no share by more than six points; the oracle variant is a maximum over features and is therefore reported at a calibrated threshold; the minimum detectable extent is the median trial of a level, a 50 %-power quantity, which is why the decision table demands a detectable share as well, and it is known only as a bracket on three inter-turn levels, so that a requirement between two levels cannot be assessed and the continuous SDR and its ratio remain the better basis for comparing alternatives (a regression of the SDR on the extent would give a continuous estimate at the price of a model for the signature law); the required share of 0.9 and the confidence of 0.95 are choices of the requirement, not of the method. Other nuisance transients (load steps, grid events) were not in the data. Finally, the SDR characterises a short-horizon change detector; the between-trial null shows that a monitor referenced to an old baseline faces a null 1.1 to 4.6 times wider on the key features, and every MDE in this paper should be read with that qualification.

7. Conclusions

A detector-free measure of fault detectability has been proposed, the signal-to-drift ratio: a scale-free statistic calibrated on each machine's own healthy non-stationarity, from which a minimum detectable extent and a detectable share above a required extent can be computed for every combi-

nation of machine type, sensor suite and failure-mode class, with a reliability screen, bootstrap intervals and a decision table over the required extent and the false-alarm level. A six-step procedure with default rules for its framing decisions turns these into the selection of the cheapest adequate sensor suite for each machine type, a comparison of the machine types on that basis, and a transferability specification for a detector that must serve several of them.

Applied to three types of wind-turbine generator subjected to the same tapped winding short-circuits on open benchmark data, the method showed that the same design-level fault is not the same physical fault across the machines, and that the faulted phase rides on the nominal extent axis unless sequence quantities are taken as phasor residuals; that where a fault becomes visible depends on the machine–drive combination, the two machines on one bench being indistinguishable through the terminal voltage and differing through the stator current by a quieter null and through the controller by both signature and null; that the cheapest sensor suite meeting a requirement depends on the alternative and on the requirement, the middle tested inter-turn level being met by drive-internal quantities on all three machines and the smallest inter-winding level requiring voltage transducers on the permanent-magnet machine alone; and that a learned detector transfers across the three machines once its features are referenced to healthy behaviour of the target machine, best to each trial's own preceding healthy interval. The findings carry tap-pair-level uncertainty and the sampling variability of the null, are robust in their orderings to the method's tuning, and are limited by one physical unit per alternative, laboratory scale, resistor-limited fault emulation and a single failure-mode class.

Three extensions follow: the design-stage instantiation, in which simulated faulted trials are combined with measured healthy nulls of hardware of the same class and validated on a few physical trials, so that the method applies before hardware exists and machine-level differences become population-level statements; other failure-mode classes and observables—bearing and rotor faults with vibration and flux observables, zero-sequence and Park's-vector features for the winding faults studied here—and a mechanical or electrochemical family, to test the claim to generality; and a cost model including integration, maintenance and missed detection in place of the ordinal ranking of suites, so that the co-selection can be posed as an optimisation. The released code, which reproduces every result from the public data, is the starting point for all three.

CRediT authorship contribution statement

Hüseyin Tayyer Canseven: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The author declares that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All data are public under the CC BY 4.0 licence: the PMSG, SCIG and synchronous-generator datasets of the InnovaPower MitDev-Eletrica project, https://github.com/InnovaPower/MitDev-Eletrica, archived on Zenodo at https://doi.org/10.5281/zenodo.15741561, https://doi.org/10.5281/zenodo.17161986 and https://doi.org/10.5281/zenodo.13685630. The complete analysis pipeline, with the derived per-trial tables, is released under an open licence at https://github.com/htcanseven/Claude (directory `scripts/`) and reproduces every number in this paper from the public data; an archived release with a DOI will be deposited on Zenodo at acceptance.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used a large language model (Claude, Anthropic) under their direction to assist with code development, literature retrieval, data analysis, and drafting and editing of the text. After using this tool, the author reviewed and edited the content as needed, verified all analyses, checked every reference against its published record, and takes full responsibility for the content of the publication. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A. Feature definitions, computation details and additional tables

This appendix defines the features (Appendix A.1), lists the computation details (Appendix A.2) and gives the sensor-suite results for the three alternatives (Table A.1), the sensitivity of the design quantities to the window length and the null quantile (Table A.2), the commissioning-baseline null (Table A.3) and the transfer results between the three alternatives (Table A.5).

Appendix A.1. Feature definitions

Let x_a, x_b, x_c be the sampled phase quantities of a window of N samples at sampling frequency f_s, w the Hann window and f_e the electrical frequency of the window, estimated from the slope of the unwrapped phase of the current space vector $\frac{2}{3}(i_a + \mathbf{a}\, i_b + \mathbf{a}^2 i_c)$, $\mathbf{a} = e^{j2\pi/3}$, whose sign gives the phase rotation (outside 0.5–1.5 times the nominal value the nominal value is used). The complex fundamental amplitude of a signal x at frequency f is $X(f) = \frac{2}{\sum_n w_n} \sum_{n=0}^{N-1} x_n\, w_n\, e^{-j2\pi f n/f_s}$. Sequence components are $X_+ = (X_a + \mathbf{a}X_b + \mathbf{a}^2 X_c)/3$ and $X_- = (X_a + \mathbf{a}^2 X_b + \mathbf{a}X_c)/3$ for an *abc* rotation (interchanged for *acb*), evaluated at $f = f_e$ for currents ($I_1 = |I_+|$, $I_2 = |I_-|$), terminal voltages (V_1, V_2) and mean-removed duty cycles (D_1, D_2); $|X(f)|$ is a peak amplitude and is divided by $\sqrt{2}$ wherever it is compared with an RMS value. The phase-referenced negative-sequence ratio of a window is the complex number $\bar{X}_2 = X_- \overline{X_+}/|X_+|^2$, whose angle is that of the negative-sequence component relative to the positive-sequence one and which therefore does not rotate with the window start; the interval mean of a residual feature is the complex mean of $\bar{X}_2$ over the windows of the interval, and its relative changes

Table A.1: Sensor suites for the three alternatives, three-cycle windows, common feature set, $\alpha = 0.95$

Suite	Machine	Feature	MDE	DS	DS nested	FA	Oracle DS	Oracle FA
Inter-turn faults								
drive	PMSG	$\lvert\Delta\bar{D}_2\rvert/D_1$	(3.0, 7.5]	76	76	0/9	68	0
drive	SCIG	$\lvert\Delta\bar{D}_2\rvert/D_1$	(3.0, 7.5]	82	82	1/9	69	0
drive	WFSG	$V_q^{\mathrm{cmd}}\,2f_e$	$\leq$11.8[a]	100	100	–	93	–
3CT	PMSG	$\lvert\Delta\bar{I}_2\rvert/I_1$	(3.0, 7.5]	70	70	0/9	63	0
3CT	SCIG	$\lvert\Delta\bar{I}_2\rvert/I_1$	(3.0, 7.5]	82	82	0/9	78	11
3CT	WFSG	$\lvert\Delta\bar{I}_2\rvert/I_1$	$\leq$11.8[a]	100	100	–	87	–
3CT+3VT	PMSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	$\leq$3.0	90	90	1/9	76	0
3CT+3VT	SCIG	$\lvert\Delta\bar{I}_2\rvert/I_1$	(3.0, 7.5]	82	82	0/9	77	11
3CT+3VT	WFSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	$\leq$11.8[a]	100	100	–	92	–
3CT+3VT+drv	PMSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	$\leq$3.0	90	90	1/9	71	0
3CT+3VT+drv	SCIG	$\lvert\Delta\bar{D}_2\rvert/D_1$	(3.0, 7.5]	82	81	1/9	72	11
3CT+3VT+drv	WFSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	$\leq$11.8[a]	100	100	–	97	–
exc	WFSG	$I_f\,2f_e$	$\leq$11.8[a]	100	100	–	87	–
Inter-winding faults								
drive	PMSG	$\lvert\Delta\bar{D}_2\rvert/D_1$	$\leq$2.8	93	87	0/9	88	0
drive	SCIG	$I_d\,2f_e$	$\leq$2.8	100	100	0/9	100	0
drive	WFSG	$V_q^{\mathrm{cmd}}\,2f_e$	$\leq$2.3	100	100	–	100	–
3CT	PMSG	$\lvert\Delta\bar{I}_2\rvert/I_1$	(2.8, 10.1]	86	86	0/9	89	0
3CT	SCIG	I_2/I_1	$\leq$2.8	98	98	0/9	97	11
3CT	WFSG	I_2/I_1	$\leq$2.3	100	100	–	100	–
3CT+3VT	PMSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	$\leq$2.8	100	100	1/9	98	0
3CT+3VT	SCIG	V_2/V_1	$\leq$2.8	100	98	2/9	98	11
3CT+3VT	WFSG	V_2/V_1	$\leq$2.3	100	100	–	100	–
3CT+3VT+drv	PMSG	$\lvert\Delta\bar{V}_2\rvert/V_1$	$\leq$2.8	100	100	1/9	96	0
3CT+3VT+drv	SCIG	$I_d\,2f_e$	$\leq$2.8	100	100	0/9	100	11
3CT+3VT+drv	WFSG	V_2/V_1	$\leq$2.3	100	100	–	100	–
exc	WFSG	$I_f\,2f_e$	$\leq$2.3	100	100	–	100	–

Fixed feature: the suite's screened feature with the smallest MDE, then the highest share, then the highest median SDR. MDE bracket (%) on the physical levels at the fault impedance matching the PMSG/SCIG protocol (WFSG: 198 trials, four tap pairs per class). DS: detectable share (%) of the fixed feature; nested: fixed feature chosen on the other tap pairs; FA: its false alarms on the nine healthy trials; oracle: per-trial best feature at the calibrated threshold, with its false-alarm rate (%; the WFSG has no healthy trials).

[a] The smallest inter-turn extent tested on the WFSG at that impedance is 11.6 %, so the value is an upper bound.

are $|\bar{X}_{2,F} - \bar{X}_{2,R_2}|/|\bar{X}_{2,R_2}|$ under fault and $|\bar{X}_{2,R_2} - \bar{X}_{2,R_1}|/|\bar{X}_{2,R_1}|$ in the null, Eq. (1) applied to the complex mean. A window belongs to R_1 when it ends before the split time and to R_2 when it starts after it, so the two halves share no samples. The RMS unbalance is $\max_k |I_{\mathrm{rms},k} - \bar{I}_{\mathrm{rms}}|/\bar{I}_{\mathrm{rms}}$ over the three phases; the harmonic ratios are $|I_a(kf_e)|/|I_a(f_e)|$ for $k = 3, 5, 7$ and the THD

Table A.2: Sensitivity of MDE/DS to the window length (3, 5, 8 cycles) and to the null quantile (α 0.90, 0.99)

Class	Feature	Machine	3 c	5 c	8 c	α 0.90	α 0.99
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	PMSG	3.0/90	3.0/88	3.0/85	3.0/92	7.5/78
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	SCIG	7.5/78	7.5/81	7.5/79	3.0/84	7.5/68
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	PMSG	7.5/70	7.5/70	7.5/70	7.5/71	7.5/61
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	SCIG	7.5/82	3.0/85	3.0/85	3.0/87	7.5/78
IT	$I_d\,2f_e$	PMSG	11.8/28	11.8/24	–/15[a]	11.8/30	–/12[a]
IT	$I_d\,2f_e$	SCIG	7.5/73	7.5/76	7.5/73	3.0/82	7.5/69
IT	$\mathrm{PI}_d\,2f_e$	PMSG	11.8/28	11.8/24	–/15[a]	11.8/30	–/11[a]
IT	$\mathrm{PI}_d\,2f_e$	SCIG	7.5/73	7.5/76	7.5/73	7.5/82	7.5/69
IT	$V_q\,2f_e$	PMSG	11.8/53	11.8/51	7.5/52	7.5/61	11.8/47
IT	$V_q\,2f_e$	SCIG	7.5/59	7.5/60	7.5/59	7.5/62	7.5/59
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	PMSG	2.8/100	2.8/100	2.8/100	2.8/100	2.8/99
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	SCIG	2.8/100	2.8/100	2.8/100	2.8/100	2.8/99
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	PMSG	10.1/86	10.1/86	10.1/86	10.1/88	10.1/85
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	SCIG	2.8/98	2.8/99	2.8/99	2.8/100	2.8/97
IW	$I_d\,2f_e$	PMSG	14.6/70	14.6/69	21.9/64[a]	14.6/72	21.9/63[a]
IW	$I_d\,2f_e$	SCIG	2.8/100	2.8/100	2.8/100	2.8/100	2.8/100
IW	$\mathrm{PI}_d\,2f_e$	PMSG	14.6/70	14.6/69	21.9/64[a]	14.6/72	21.9/63[a]
IW	$\mathrm{PI}_d\,2f_e$	SCIG	2.8/100	2.8/100	2.8/100	2.8/100	2.8/100
IW	$V_q\,2f_e$	PMSG	10.1/89	10.1/89	10.1/87	2.8/91	10.1/82
IW	$V_q\,2f_e$	SCIG	10.1/88	10.1/88	10.1/86	2.8/94	10.1/84

Cells: MDE (%; upper end of the bracket on the physical levels; '–': largest level not detectable) / DS (%). Window-length columns at $\alpha = 0.95$; quantile columns at five cycles. IT: inter-turn; IW: inter-winding.

[a] The feature fails the reliability screen at that setting.

is $\sqrt{\sum_{k=2}^{20} |I_a(kf_e)|^2}/|I_a(f_e)|$. For the synchronous-frame currents i_d, i_q and the torques the features are the standard deviation of the mean-removed signal and its amplitudes $|Y(f_e)|$ and $|Y(2f_e)|$; for the PI outputs the standard deviation and $|Y(2f_e)|$; for the measured and commanded synchronous-frame voltages $|Y(2f_e)|$; for the field current the standard deviation and $|Y(2f_e)|$; speed and DC-link features are standard deviations. Windows are n_c electrical cycles long with a hop of one cycle, and a feature is undefined for a trial whose faulted interval holds no complete window. On the WFSG bench the measured converter-side phase voltages serve as terminal voltages, the synchronous-frame voltages are obtained from them with the encoder angle, the commanded voltages are the controller's, and the electrical torque is computed from power and speed.

Table A.3: Within-trial versus commissioning-baseline null, five-cycle windows, $\alpha = 0.95$

Class	Feature	M.	q_{95}		Median SDR		DS	
			within	baseline	within	baseline	within	baseline
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	P	0.123	0.382	6.84	2.36	88	72
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	P	0.123	0.382	52.7	14.3	100	92
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	P	0.242	0.458	1.7	0.937	70	49
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	P	0.242	0.458	13.7	6.95	86	85
IT	$\mathrm{PI}_d\,2f_e$	P	0.781	0.870	0.37	0.441	24	20
IW	$\mathrm{PI}_d\,2f_e$	P	0.781	0.870	3	2.4	69	67
IT	$V_q\,2f_e$	P	0.197	0.277	1.13	0.872	51	44
IW	$V_q\,2f_e$	P	0.197	0.277	18.7	10.7	89	81
IT	$\lvert\Delta\bar{V}_2\rvert/V_1$	S	0.051	0.159	6.9	2.36	81	67
IW	$\lvert\Delta\bar{V}_2\rvert/V_1$	S	0.051	0.159	39	13.5	100	93
IT	$\lvert\Delta\bar{I}_2\rvert/I_1$	S	0.046	0.146	6.34	2.05	85	69
IW	$\lvert\Delta\bar{I}_2\rvert/I_1$	S	0.046	0.146	45.8	13.9	99	89
IT	$\mathrm{PI}_d\,2f_e$	S	0.017	0.078	6.18	1.29	76	53
IW	$\mathrm{PI}_d\,2f_e$	S	0.017	0.078	68.4	16	100	88
IT	$V_q\,2f_e$	S	0.066	0.131	2.34	1.54	60	58
IW	$V_q\,2f_e$	S	0.066	0.131	22.5	11.2	88	86

Four key features. Within: reference immediately before the fault; baseline: the healthy trial at the same operating point, recorded at another time. DS: detectable share (%). IT: inter-turn; IW: inter-winding; P: PMSG; S: SCIG.

Table A.4: Null quantile q_{95} of seven features with its 95 % sampling interval, five-cycle windows

Feature	PMSG q_{95} [CI]	$q_{95}/1.96\sigma$	SCIG q_{95} [CI]	$q_{95}/1.96\sigma$
V_2/V_1	0.107 [0.087, 0.115]	1.03	0.037 [0.034, 0.045]	0.95
$\lvert\Delta\bar{V}_2\rvert/V_1$	0.123 [0.114, 0.146]	–	0.051 [0.045, 0.059]	–
I_2/I_1	0.172 [0.148, 0.205]	1.00	0.038 [0.034, 0.045]	1.06
$\lvert\Delta\bar{I}_2\rvert/I_1$	0.242 [0.211, 0.277]	–	0.046 [0.039, 0.053]	–
$I_d\,2f_e$	0.774 [0.629, 0.918]	1.11	0.017 [0.015, 0.021]	0.95
$\mathrm{PI}_d\,2f_e$	0.781 [0.618, 0.858]	1.12	0.017 [0.015, 0.021]	0.95
$V_q\,2f_e$	0.197 [0.166, 0.217]	1.06	0.066 [0.060, 0.086]	1.03

Interval from a cluster bootstrap over the 24 tap pairs and the healthy set of each machine (2000 resamples). $q_{95}/1.96\sigma$: the quantile relative to the Gaussian value for the same standard deviation of the signed null; not applicable to the residual features, whose null is a magnitude.

Appendix A.2. Computation details

Intervals. PMSG and SCIG: reference interval 0.30–0.98 s after the start of each record, split at 0.62 s into R_1 (0.32 s) and R_2 (0.36 s); contactor

Table A.5: Transfer between the three alternatives, self-referenced GBDT detector, three-cycle windows, out-of-reference windows

Trained on	Tested on	AUC [CI]	TPR [CI]	FA healthy (%)
PMSG	PMSG	0.90 [0.85, 0.94]	0.71 [0.56, 0.85]	3
SCIG	SCIG	0.93 [0.89, 0.96]	0.77 [0.64, 0.89]	1
WFSG	WFSG	0.97 [0.94, 1.00]	0.93 [0.85, 0.99]	–
PMSG	SCIG	0.89 [0.83, 0.95]	0.67 [0.53, 0.82]	0
PMSG	WFSG	0.98 [0.96, 1.00]	0.94 [0.86, 0.99]	–
SCIG	PMSG	0.85 [0.79, 0.91]	0.41 [0.25, 0.54]	0
SCIG	WFSG	0.98 [0.95, 0.99]	0.84 [0.72, 0.95]	–
WFSG	PMSG	0.86 [0.80, 0.92]	0.63 [0.48, 0.77]	0
WFSG	SCIG	0.90 [0.85, 0.94]	0.57 [0.44, 0.74]	0
SCIG+WFSG	PMSG	0.87 [0.81, 0.93]	0.55 [0.39, 0.71]	0
PMSG+WFSG	SCIG	0.90 [0.84, 0.95]	0.68 [0.53, 0.82]	0
PMSG+SCIG	WFSG	0.99 [0.97, 1.00]	0.95 [0.86, 0.99]	–

Features $|z|$-referenced to the first half of each trial's own reference interval. AUC and TPR at 1 % FPR on the out-of-reference windows with 95 % intervals from a bootstrap over the tested machine's trials; FA: share of the healthy trials' fault-time windows above the threshold giving 1 % FPR on all negatives. The WFSG has no healthy trials, so its negatives are reference and recovery windows only and its rows are optimistic.

commanded at 1.000 s; faulted interval from 30 ms after the command to 10 ms before its release; the relay picks up 85–90 ms after the command, so the faulted interval holds about 55 ms of healthy time (at most 15 % of its length), which dilutes the fault-interval mean and makes the PMSG/SCIG SDRs slightly conservative; recovery windows start 0.30 s after the release. WFSG: fault interval located from the measured fault current as the first contiguous interval in which its 10 ms RMS exceeds $\max(0.3\,\mathrm{A}, 6\sigma)$, σ being the standard deviation of that RMS over 0.05–0.40 s, ending at the first fall below the threshold after 50 ms; reference interval from 0.05 s after the start of the record to 10 ms before the onset, split at 0.30 s; recovery windows start 0.15 s after the end of the fault interval. At 23.9–63.7 Hz a WFSG faulted interval holds between zero and eleven three-cycle windows; four trials hold none and are dropped, and 81 of 438 hold fewer than three (all 50 trials at 1671 rpm, 26 of 206 at 1800 rpm, 4 at 716 and 1 at 955 rpm).

Null and screen. The null is pooled over all 225 records of each of the PMSG and SCIG and over the 438 inter-turn and inter-winding records of the WFSG. The screen flags a feature when its count of false alarms on the nine healthy trials is incompatible with the nominal rate $1 - \alpha$ at the 5 % level (one-sided binomial test: three or more at $\alpha = 0.95$, two or more at

$\alpha = 0.99$, four or more at $\alpha = 0.90$; with nine trials the actual size of the rule is 0.8, 0.4 and 0.8 %) and excludes features whose null quantile exceeds one or is unbounded. The mechanical-torque features of the SCIG are excluded at every setting, because their healthy level crosses zero and the relative null is unbounded. Beyond them, at $\alpha = 0.95$ the screen removes D_2/D_1 on the SCIG at three and five cycles (three false alarms) and $T_e\,1f_e$ on the SCIG at five cycles (null quantile 1.01); at eight cycles $T_e\,1f_e$ and $2f_e$ on the SCIG and nine features on the PMSG whose null exceeds one ($I_d\,1f_e$ and $2f_e$, $\mathrm{PI}_d\,2f_e$, T_e and T_m at $1f_e$ and $2f_e$, $V_d\,2f_e$ and $V_d^{\mathrm{cmd}}\,2f_e$). At $\alpha = 0.99$ and five cycles the null-quantile rule removes ten PMSG features, among them the d-axis features, and three SCIG features; at $\alpha = 0.90$ the binomial rule removes speed std, I_q std and PI_d std on the PMSG and D_2/D_1 and V_{dc} std on the SCIG (four false alarms each).

Bootstrap. 2000 resamples, seed 1. Median SDR and detectable share of a class: cluster bootstrap over the twelve tap pairs, the same resampled pairs for both machines; in every resample the null quantile is re-estimated from the resampled pairs of the class together with a cluster resample of the remaining thirteen clusters of the machine (the other class's pairs and the healthy set). Minimum detectable extent: operating-point trials resampled within each physical level, the null re-estimated from a cluster resample of all 25 clusters, and the limit-of-detection rule applied to each resample. The sampling interval of the null quantile itself (Table A.4) comes from the same cluster resampling. Decision table: 1000 resamples per suite, alternative, class and α (seed 7); in every resample the operating-point trials are resampled within each level, the null is re-estimated, the fixed feature is re-selected by the rule of Section 3.2 (smallest MDE, then highest share, then highest median SDR), and the joint event $\{\mathrm{MDE} \le e_{\mathrm{req}}$ and $\mathrm{DS}(e \ge e_{\mathrm{req}}) \ge 0.9\}$ is scored at every tested level e_{req} of the common bench; a suite inherits a strictly higher probability of a cheaper subset; *meets* at probability ≥ 0.95, *fails* at ≤ 0.05; a requirement below the smallest or above the largest extent tested on a machine is not testable.

Transfer. Models: scikit-learn histogram gradient boosting (maximum depth 3, 200 iterations, learning rate 0.08, random state 0) and logistic regression with standardisation ($C = 1$, 3000 iterations); five-fold GroupKFold with one group per tap pair and machine, healthy trials grouped by speed. Inputs are the window features, the phasor residuals entering through the real and imaginary parts of $\bar{X}_2$. Four referencing rules: physical units; z-scores on all healthy trials of the target machine; z-scores on the healthy trial recorded

at the same operating point (the commissioning baseline of Table A.3); and the absolute z-score against the windows of the trial's own reference interval that end before 0.62 s (PMSG/SCIG) or 0.30 s (WFSG), the normalising set. Metrics: area under the receiver operating characteristic curve (AUC) and true-positive rate at 1 % false-positive rate, on all windows and on the windows outside the normalising set, with 95 % percentile intervals from 500 cluster resamples of the tested machine's trials (tap pairs and healthy speed groups), and the false-alarm rate of the all-negatives 1 % threshold on the fault-time windows of the healthy trials.

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